Pointwise Complexity for Gaussian Fields: Upper Envelopes, Algorithmic Lower Envelopes, and Finite-Scale Renormalization

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Abstract

We prove a variance-aware pointwise majorizing-measure theorem for centered Gaussian processes. Classical generic chaining characterizes the scalar quantity $\mathbb{E}\sup_{x\in T} X_x$; the theorem here gives a simultaneous high-probability envelope for the entire field. For an ambient prior μ, the envelope at x is governed by a localized Fernique–Talagrand functional $\Phi_\mu(x)$, truncated at the variance scale $\sigma(x) = (\mathbb{E}X_x^2)^{1/2}$, together with the corresponding Gaussian tail term. Optimizing the resulting in-expectation bound recovers the anchored Fernique–Talagrand scale up to universal constants. The theorem is intended as a standard reference whenever one needs a field-level refinement of classical generic chaining, and as the Gaussian-process counterpart of pointwise empirical-process bounds for deep neural networks (Li and Xu, 2026).

We also derive a one-scale Bayesian algorithmic lower envelope from the interactive Fano/data-processing principle (Chen et al., 2024). For a known prior, an observation channel, and a concrete estimator $\widehat{t}(Y)$, the lower bound is expressed through the exact ghost small-ball mass $\mathbb{E}_{Y\sim Q}\pi(B_d(\widehat{t}(Y),\Delta))$, rather than a worst-case covering number. In Gaussian location experiments, nearest-neighbor decoding first yields a Bayes lower bound on location error. The nearest-neighbor inequality then converts this error into a lower bound on the range of the associated Gaussian process, valid for arbitrary priors. A subsequent fractional-covering relaxation recovers a one-scale Sudakov-type lower bound for priors localized at the tested scale. Together, these results show that Bayesian algorithmic lower bounds can serve as local-geometric certificates of pointwise-complexity optimality for fixed algorithms across different and potentially unknown priors.

We further isolate a finite-dimensional pointwise-complexity principle for renormalization maps. This part is deliberately finite-cutoff: it is not a construction of a continuum interacting quantum field. Its purpose is to make precise the structural analogy with nonlinear feature learning in deep networks (Li and Xu, 2026). An exact finite-scale telescoping identity replaces one RG scale at a time; the resulting secant response Gram defines a pointwise ellipsoidal metric; and a data-independent hierarchical prior over phase labels and active subspaces splits the complexity into a local effective dimension and a global atlas cost. This finite-scale theorem and a nonconvex phase-atlas example clarify why a useful nonlinear covering theory for interacting fields must first expose the correct response Gram and the correct atlas, rather than applying a scalar Lipschitz or global-supremum bound to the whole space.

Finally, we apply the Gaussian theorem to finite pinned Gaussian free fields through the second Ray–Knight isomorphism. The resulting target-set local-time envelopes separate direct field-level control from reductions through full cover time. On phase-star graphs, a selected subatlas pays its own atlas code length and chart cost, while any full-cover route must pay for all phases.

Contents

1 Introduction and Main Results **2**
1.1 Main pointwise Gaussian theorem . . . 3
1.2 Bayesian algorithmic lower bounds and Sudakov-type comparison . . . 5
1.3 Main finite-scale renormalization theorem . . . 9
1.4 Main local-time application: target fields versus cover time . . . 10
1.5 Organization . . . 10

2 Proofs of the Gaussian and Algorithmic Lower Results **10**
2.1 Proof of the Pointwise Gaussian Theorem . . . 10
2.2 Proofs for the Bayesian algorithmic lower bounds . . . 14

3 Finite-Scale Renormalization and Pointwise Dimension **15**
3.1 Exact finite-scale telescoping for nonlinear RG maps . . . 16
3.2 Hierarchical phase/subspace prior . . . 19
3.3 Free Gaussian RG as the fixed-kernel baseline . . . 20
3.4 A finite nonconvex phase-atlas example . . . 20
3.5 Relation to rigorous RG, ADC, universal approximation, and Yang–Mills . . . 22

4 Graph Local-Time Applications: Target Fields versus Cover Time **24**
4.1 Graph setup, pinned GFF, and Ray–Knight . . . 24
4.2 Target local times versus cover-time reduction . . . 25

5 Scope and Outlook **31**

1 Introduction and Main Results

Classical generic chaining controls the scalar quantity $\mathbb{E}\sup_{x\in T} X_x$ for a Gaussian process $\{X_x\}_{x\in T}$. For many field-level questions, however, this scalar is not the first object one wants to estimate. In the graph setting, the second Ray–Knight theorem identifies a local-time field at an inverse-local-time stopping rule with a shifted square of a Gaussian free field. In constructive and lattice field theory, one similarly studies local observables before passing to global maxima or continuum limits. This suggests that the Gaussian input should itself be pointwise: one should first prove a simultaneous field-level envelope and only afterwards take suprema or threshold events.

A further motivation comes from recent work on pointwise generalization for deep neural networks (Li and Xu, 2026). There the central object is not a class-wide supremum but a hypothesis-dependent finite-scale complexity. The proof is structural: an exact non-perturbative telescoping expansion exposes learned feature Gram matrices, and a hierarchical local-chart/global-atlas prior converts the resulting data-dependent active subspaces into a data-independent bound. The present paper develops the Gaussian-process counterpart of that viewpoint. Theorem 1.1 can serve as a standard reference whenever one needs a field-level refinement of classical generic chaining.

The paper then records a finite-cutoff renormalization analogue of this pointwise viewpoint. The analogy with deep feature learning is not merely that layers resemble scales. The shared proof architecture is exact finite-scale telescoping, a pointwise ellipsoidal metric induced by a learned or renormalized response Gram, and a hierarchical atlas separating local effective dimension from global prior-independence cost. We state this part as an abstract deterministic complexity theorem, together with a chartwise nonconvex example. The graph application later in the paper uses the same finite-scale logic to compare two probabilistic questions: controlling a local-time field on a specified target set versus reducing the problem to global cover time. The phase-star example shows that retaining the target field can give a strictly smaller scale than any argument that first proves full cover.

We also record a one-scale Bayesian lower-envelope counterpart. It is not a pathwise lower bound on each Gaussian coordinate. Rather, it is a data-processing Bayes-risk lower bound for a specified prior, observation channel, and estimator. In Gaussian location experiments, nearest-neighbor decoding converts this Bayes location error into a lower bound on the Gaussian range. This conversion is variance-normalized and valid for arbitrary priors. When applied to hard priors localized at the tested radius, it yields a Sudakov-type scale. This provides a simpler, one-scale, and prior-agnostic route to understanding the more involved and inherently multiscale Sudakov proof. This separates the frequentist, simultaneous, multiscale upper envelope from the Bayesian, one-scale, estimator-dependent lower envelope. Because sharp matching between upper and lower bounds is often out of reach for general rich classes, the alignment between frequentist pointwise upper bounds and Bayesian algorithmic lower bounds provides a local-geometric route to certifying the practical optimality of fixed algorithms and estimators.

We keep the limitations explicit. Pointwise complexity can organize finite-cutoff Gaussian and RG geometry, but it does not by itself prove Wick factorization, marginal triviality, a universal-approximation-based RG proof, or the Yang–Mills mass gap. Those problems require substantial model-specific contraction, Gaussianization, or gauge-invariant correlation estimates.

1.1 Main pointwise Gaussian theorem

Let $\{X_x\}_{x\in T}$ be a centered Gaussian process on a finite index set T. Fix an anchor $x_0 \in T$ such that $X_{x_0} = 0$ almost surely, and define

$$d(x,y) := \big(\mathbb{E}|X_x - X_y|^2\big)^{1/2}, \qquad \sigma(x) := (\mathbb{E}X_x^2)^{1/2} = d(x,x_0), \qquad R_* := \sup_{x\in T} \sigma(x).$$

For an ambient prior $\mu \in \Delta(T)$, where $\Delta(T)$ denotes the set of probability measures on T, define the localized pointwise Fernique–Talagrand functional

$$\Phi_\mu(x) := \int_0^{4\sigma(x)} \sqrt{\log \frac{1}{\mu(B_d(x,\varepsilon))}}\, d\varepsilon, \qquad B_d(x,\varepsilon) := \{y \in T : d(x,y) \le \varepsilon\}. \tag{1.1}$$

Here the upper limit is local: it is proportional to the standard deviation $\sigma(x)$, rather than the global radius. The factor 4 is inessential; it absorbs the two constant-factor losses coming from anchoring and from the nearest-point pushforward used in the ambient-equivalence step. If one instead works with the non-local full-radius integral, then the upper limits R_* and $2R_*$ are equivalent up to universal constants, since the entropy integrand is nonincreasing in ε.

Write

$$\Phi_{\mu,*} := \sup_{x\in T} \Phi_\mu(x), \qquad \log_2^+(u) := \max\{\log_2 u, 0\}.$$

The degenerate case $R_* = 0$ is trivial, and the statements below are understood to be vacuous if $\Phi_{\mu,*} = +\infty$.

For $r_0 > 0$, $s_0 \in (0, R_*]$, and $\delta \in (0,1)$, set

$$M_{r_0,s_0} := \big(1 + \lceil \log_2^+(\Phi_{\mu,*}/r_0)\rceil\big)\big(1 + \lceil \log_2(R_*/s_0)\rceil\big), \tag{1.2}$$

and

$$\mathcal{E}_{\mu,r_0,s_0}(x;\delta) := \max\{\Phi_\mu(x), r_0\} + \max\{\sigma(x), s_0\}\sqrt{\log\left(\frac{eM_{r_0,s_0}}{\delta}\right)}. \tag{1.3}$$

Thus all logarithmic losses from the two successive pointwise-peeling steps are isolated in M_{r_0,s_0}.

We state the main result as a variance-aware pointwise Gaussian majorizing-measure theorem in high probability.

Theorem 1.1 (Pointwise Gaussian majorizing-measure bound)**.** *Assume $0 < R_* < \infty$ and $\Phi_{\mu,*} < \infty$. There exists an absolute constant $C > 0$ such that, for every ambient prior $\mu \in \Delta(T)$, every $r_0 > 0$, every $s_0 \in (0, R_*]$, and every $\delta \in (0,1)$, with probability at least $1-\delta$,*

$$\forall x \in T: \qquad X_x \le C\,\mathcal{E}_{\mu,r_0,s_0}(x;\delta). \tag{1.4}$$

Consequently, applying the same bound to X and $-X$ and taking a union bound, with probability at least $1-\delta$,

$$\forall x \in T: \qquad |X_x| \le A_{\mu,r_0,s_0}(x;\delta), \qquad A_{\mu,r_0,s_0}(x;\delta) := C\,\mathcal{E}_{\mu,r_0,s_0}(x;\delta/2). \tag{1.5}$$

When r_0, s_0 are fixed, we abbreviate A_{μ,r_0,s_0} to A_μ.

Before presenting the in-expectation consequence, we record the corresponding anchored form of the classical generic-chaining scale. This is a localized reformulation of the usual Fernique–Talagrand theorem: the cutoff $4\sigma(x)$ is not the standard full-radius presentation, and therefore the global radius R_* must remain as part of the anchored scale.

Proposition 1.2 (Classical scale in anchored local form)**.** *There exist absolute constants $0 < c < C < \infty$ such that*

$$c\left(R_* + \inf_{\mu\in\Delta(T)} \Phi_{\mu,*}\right) \le \mathbb{E}\sup_{x\in T} X_x \le C\left(R_* + \inf_{\mu\in\Delta(T)} \Phi_{\mu,*}\right). \tag{1.6}$$

Moreover, since the process is centered and anchored,

$$\mathbb{E}\sup_{x\in T}|X_x| \asymp \mathbb{E}\sup_{x\in T} X_x.$$

Therefore

$$\mathbb{E}\sup_{x\in T}|X_x| \asymp R_* + \inf_{\mu\in\Delta(T)} \Phi_{\mu,*}. \tag{1.7}$$

We now record the in-expectation consequence of Theorem 1.1. In particular, after optimizing over the ambient prior, the resulting global in-expectation bound is sharp up to universal constants. The variance-radius term is not an additional loss: it is part of the anchored generic-chaining scale in (1.7), while the peeling logarithms disappear by taking $r_0 \ge \Phi_{\mu,*}$ and $s_0 = R_*$.

Corollary 1.3 (In-expectation consequence and sharp global scale)**.** *Under the assumptions of Theorem 1.1, there exists an absolute constant $C > 0$ such that*

$$\mathbb{E}\left[\sup_{x\in T}\frac{\big(|X_x| - C\max\{\Phi_\mu(x), r_0\}\big)_+}{\max\{\sigma(x), s_0\}}\right] \le C\sqrt{\log(eM_{r_0,s_0})}. \tag{1.8}$$

Consequently, for every $x \in T$,

$$\mathbb{E}|X_x| \le C\max\{\Phi_\mu(x), r_0\} + C\max\{\sigma(x), s_0\}\sqrt{\log(eM_{r_0,s_0})}. \tag{1.9}$$

At the global level, taking $r_0 \geq \Phi_{\mu,*}$ and $s_0 = R_*$ gives $M_{r_0,s_0} = 1$, hence

$$\mathbb{E} \sup_{x \in T} |X_x| \leq C(r_0 + R_*). \tag{1.10}$$

Letting $r_0 \downarrow \Phi_{\mu,*}$, then optimizing over μ, and using (1.7), we obtain the sharp equivalence

$$\mathbb{E} \sup_{x \in T} |X_x| \asymp R_* + \inf_{\mu \in \Delta(T)} \Phi_{\mu,*}. \tag{1.11}$$

Thus the optimized expectation version is optimal up to absolute constants.

Remark (variance term versus chaining complexity). The term involving $\sigma(x)$ is a variance-sensitive Gaussian tail term. It is necessary for high-probability control, already for a single non-anchor point $X_x \sim N(0, \sigma(x)^2)$, but it is not a replacement for the chaining complexity $\Phi_\mu(x)$, which carries the local geometry of the index set.

For a fixed marginal,

$$\mathbb{E}|X_x| = \sqrt{\frac{2}{\pi}}\, \sigma(x),$$

so the extra factor $\sqrt{\log(eM_{r_0,s_0})}$ in (1.9) is not intrinsic to $\mathbb{E}|X_x|$. It is the price of extracting a simultaneous pointwise envelope by peeling first over the local Fernique–Talagrand complexity and then over the local variance. At the global in-expectation level, this peeling factor disappears, yielding the sharp scale in (1.11).

Remark (pointwise high probability). For Gaussian processes, subset-level in-expectation bounds transfer to high-probability bounds through Borell–Tsirelson concentration, with variance scale $\sigma_H = \sup_{x \in H} (\mathbb{E} X_x^2)^{1/2}$ for a fixed subset H. The genuinely pointwise statement is not obtained from this transfer alone: after the subset-homogeneous estimate is established, one first applies uniform pointwise peeling to localize the Fernique–Talagrand term $\Phi_\mu(x)$, and then applies the same peeling mechanism a second time to improve the tail scale from the global radius to the local variance $\sigma(x)$. Conversely, the resulting high-probability envelope implies the stated in-expectation consequence by integrating the tail.

1.2 Bayesian algorithmic lower bounds and Sudakov-type comparison

The pointwise upper envelope is frequentist, simultaneous, multiscale, and pathwise in the Gaussian field. A lower-bound analogue should therefore be stated with care. We first record a one-scale Bayesian and estimator-dependent primitive. Let (T, d) be a finite metric space, let $\pi \in \Delta(T)$ be a known prior, and let $\{P_t : t \in T\}$ be an experiment on an observation space $\mathcal{Y}$. For an estimator $\widehat{t} : \mathcal{Y} \to T$, a reference law $Q \in \Delta(\mathcal{Y})$, and a scale $\Delta > 0$, define

$$\rho_{\Delta,Q}(\widehat{t}) := \mathbb{P}_{t \sim \pi,\, Y \sim Q}\{d(t, \widehat{t}(Y)) < \Delta\} = \mathbb{E}_{Y \sim Q}\, \pi\big(B_d(\widehat{t}(Y), \Delta)\big)\,,$$

and

$$\mathcal{I}_\pi(Q) := \mathbb{E}_{t \sim \pi} D_{\mathrm{KL}}(P_t \| Q).$$

The following result is the Gaussian-process and KL specialization of the interactive Fano method of Chen et al. (2024, Theorem 2). The original theorem provides a unifying information-theoretic principle for proving lower bounds in learning and decision-making problems.

Proposition 1.4 (Bayesian algorithmic lower envelope)**.** *For every estimator $\widehat{t}$, every $\Delta > 0$, and every reference law Q with $0 < \rho_{\Delta,Q}(\widehat{t}) < 1$,*

$$\mathbb{P}_{t\sim\pi,\, Y\sim P_t}\{d(t,\widehat{t}(Y)) \geq \Delta\} \geq \left(1 - \frac{\mathcal{I}_\pi(Q) + \log 2}{\log(1/\rho_{\Delta,Q}(\widehat{t}))}\right)_+ . \tag{1.12}$$

Consequently,

$$\mathbb{E}_{t\sim\pi,\, Y\sim P_t} d(t,\widehat{t}(Y)) \geq \sup_{\Delta>0,\, Q} \Delta \left(1 - \frac{\mathcal{I}_\pi(Q) + \log 2}{\log(1/\rho_{\Delta,Q}(\widehat{t}))}\right)_+ . \tag{1.13}$$

The key point is that $\rho_{\Delta,Q}(\widehat{t})$ is not immediately replaced by the class-wide quantity

$$q_\pi(\Delta) := \sup_{a\in T} \pi(B_d(a,\Delta)).$$

Keeping the exact ghost mass $\mathbb{E}_{Y\sim Q}\pi(B_d(\widehat{t}(Y),\Delta))$ makes the lower bound algorithmic: it is attached to the image of the actual estimator under ghost data. Replacing it by $q_\pi(\Delta)$ gives the class-wide small-ball term used in classical Fano-type lower bounds (Zhang, 2006) and in the fractional-covering specialization of Chen et al. (2024). In the Gaussian nearest-neighbor comparison below, this small-ball term must still be balanced against the information radius of the Gaussian channel.

We next present a lower-bound comparison for the Gaussian process itself, rather than only for its induced distance. The result has three layers. First, the Bayesian lower envelope gives a risk lower bound for the concrete nearest-neighbor estimator. Second, the nearest-neighbor inequality converts this risk into a lower bound on a Gaussian range. Third, replacing the exact ghost mass by a class-wide small-ball mass gives the entropy term that appears in fractional-covering lower bounds. Optimizing this small-ball term over hard priors is exactly the linear-programming duality used in Chen et al. (2024); however, in the present Gaussian comparison there remains an additional information-radius term V_π. Therefore the nearest-neighbor route gives the Sudakov scale directly only for hard priors localized at the tested radius.

Theorem 1.5 (Nearest-neighbor comparison and fractional small-ball obstruction)**.** *Let $v : T \to \mathcal{H}$ be an embedding into a finite-dimensional Hilbert space and set $d(s,t) = \|v_s - v_t\|_{\mathcal{H}}$. Let*

$$Y = v_\Theta + \tau Z, \qquad \Theta \sim \pi, \qquad Z \sim N(0, I_{\mathcal{H}}),$$

and define

$$\bar{v}_\pi := \int_T v_t\, \pi(dt), \qquad \mathsf{V}_\pi := \iint d(s,t)^2\, \pi(ds)\pi(dt), \qquad Q_\tau := N(\bar{v}_\pi, \tau^2 I_{\mathcal{H}}).$$

Let $\widehat{\Theta}_{\mathrm{nn}}(Y) \in \arg\min_{a\in T} \|Y - v_a\|$ be the nearest-neighbor estimator. If, for some $c \in (0,1)$ and $\Delta > 0$,

$$\frac{\mathsf{V}_\pi}{4\tau^2} + \log 2 \leq (1-c)\log \frac{1}{\rho_{\Delta,Q_\tau}(\widehat{\Theta}_{\mathrm{nn}})}, \tag{1.14}$$

then

$$\mathbb{E}_{\Theta,Y} d(\Theta, \widehat{\Theta}_{\mathrm{nn}}(Y)) \geq c\Delta, \tag{1.15}$$

and, for the Gaussian process $X_t := \langle Z, v_t\rangle$ built from the same Gaussian vector Z,

$$\mathbb{E}_{\Theta,Z}\big[X_{\widehat{\Theta}_{\mathrm{nn}}} - X_\Theta\big] \geq \frac{c^2\Delta^2}{2\tau}. \tag{1.16}$$

Consequently,

$$\mathbb{E}\sup_{s,u\in T}(X_s - X_u) \geq \frac{c^2\Delta^2}{2\tau}. \tag{1.17}$$

Now set

$$q_\pi(\Delta) := \sup_{a\in T}\pi(B_d(a,\Delta)), \qquad L_\pi(\Delta) := \log\frac{1}{q_\pi(\Delta)}.$$

Since $\rho_{\Delta,Q_\tau}(\widehat{\Theta}_{\mathrm{nn}}) \leq q_\pi(\Delta)$, if $L_\pi(\Delta)$ is larger than a universal constant, one may choose $\tau^2 \asymp \mathsf{V}_\pi/L_\pi(\Delta)$ so that (1.14) holds with an absolute $c \in (0,1)$. This yields the variance-normalized one-scale obstruction

$$\mathbb{E}\sup_{s,u\in T}(X_s - X_u) \gtrsim \Delta^2\sqrt{\frac{L_\pi(\Delta)}{\mathsf{V}_\pi}}. \tag{1.18}$$

In particular, if the hard prior is localized at scale Δ, meaning $\mathsf{V}_\pi \leq C_0\Delta^2$, then

$$\mathbb{E}\sup_{s,u\in T}(X_s - X_u) \gtrsim_{C_0} \Delta\sqrt{L_\pi(\Delta)}. \tag{1.19}$$

Equivalently, with

$$N^{\mathrm{loc}}_{\mathrm{spread}}(T,d,\Delta;C_0) := \sup_{\pi:\,\mathsf{V}_\pi\leq C_0\Delta^2}\frac{1}{\sup_{a\in T}\pi(B_d(a,\Delta))}, \tag{1.20}$$

one obtains, whenever the logarithm is above the universal threshold required by the information condition,

$$\mathbb{E}\sup_{s,u\in T}(X_s - X_u) \gtrsim_{C_0} \Delta\sqrt{\log N^{\mathrm{loc}}_{\mathrm{spread}}(T,d,\Delta;C_0)}. \tag{1.21}$$

The preceding theorem directly recovers a Sudakov-scale lower bound under a natural localized hard-prior condition. This is the precise mild condition needed by the nearest-neighbor route: the prior used to certify small balls must also have second moment comparable to the tested radius.

Corollary 1.6 (Localized packing form)**.** *Let $S \subset T$ be 2Δ-separated and assume that, for some $t_0 \in T$ and $C_0 < \infty$,*

$$S \subset B_d(t_0, C_0\Delta).$$

If $|S| = M \geq M_0$, where M_0 is a universal constant, then

$$\mathbb{E}\sup_{s,u\in T}(X_s - X_u) \gtrsim_{C_0} \Delta\sqrt{\log M}.$$

For comparison with classical covering notation, we record the finite-set fractional-covering duality used in Chen et al. (2024). Recall that the fractional covering number is equivalent to the canonical covering number, up to absolute-constant changes in the covering radius.

Lemma 1.7 (Fractional-covering duality)**.** *For a finite metric space (T,d) and any $\Delta > 0$,*

$$N_{\mathrm{frac}}(T,d,\Delta) := \inf_{\mu\in\Delta(T)}\sup_{x\in T}\frac{1}{\mu(B_d(x,\Delta))} = \sup_{\pi\in\Delta(T)}\frac{1}{\sup_{a\in T}\pi(B_d(a,\Delta))}. \tag{1.22}$$

The equality (1.22) is an entropy duality. It optimizes the small-ball term, but it does not remove the factor V_π in (1.18), which comes from the information radius of the Gaussian location channel. There is therefore no contradiction between the fractional-covering lower bounds of Chen et al. (2024) and the localization requirement above: the fractional-covering duality identifies the optimal hard prior for the small-ball obstruction, while the nearest-neighbor Gaussian comparison also requires that this hard prior have controlled second moment at scale Δ. Thus the comparison directly gives the localized obstruction (1.21) and the localized packing form in Corollary 1.6.

The unrestricted classical Sudakov minoration is a separate Gaussian comparison theorem, not a consequence of linear-programming duality alone. Standard proofs use nontrivial Gaussian comparison and convex-geometric duality arguments; see, for example, the expositions of Pollard (2001) and Ledoux and Talagrand (1991, Chapter 3). The full lower half of the Fernique–Talagrand majorizing-measure theorem is stronger and multiscale (Talagrand, 2005). In the present paper we therefore keep the roles separate: the Bayesian algorithmic envelope supplies a one-scale, decision-theoretic, and prior-agnostic primitive, while the unrestricted Sudakov and generic-chaining lower bounds require the classical Gaussian comparison or multiscale localization machinery.

Thus Proposition 1.4 should be viewed as the decision-theoretic one-scale primitive. The exact ghost mass $\rho_{\Delta,Q}(\widehat{t})$ is the pointwise, estimator-dependent quantity. Replacing it by $q_\pi(\Delta)$ and then using Lemma 1.7 recovers the class-wide fractional-covering entropy, but the Gaussian nearest-neighbor route keeps the additional information-radius normalization displayed in (1.18). The prior is therefore used as a device for averaging and comparison; the certified object remains the performance of a concrete estimator, rather than an assertion that an unknown parameter is literally sampled from the prior.

Why keep the Bayesian and algorithmic form. The distinction clarifies the role of minimax reasoning. If the goal is Bayes risk under a particular known prior π, then a benchmark of the form

$$\inf_{\widehat{t}} \sup_{t \in T} R(t, \widehat{t})$$

may be poorly aligned with the problem: it asks for performance against the worst point of T, whereas the Bayesian problem asks for performance under the given prior geometry. Conversely, maximizing over priors can erase the structure that makes a particular algorithm appropriate. In this sense, the algorithm-dependent ghost mass in $\rho_{\Delta,Q}(\widehat{t})$ records how the actual estimator maps ghost observations into metric balls. No single algorithm can be uniformly best for all possible priors or all possible geometries, so a well-specified Bayesian lower envelope can be a more informative optimality certificate for a pointwise frequentist upper envelope than a purely class-wide minimax statement. The algorithmic form is therefore crucial: it allows us to evaluate a fixed estimator under specified prior geometries for the purpose of pointwise comparison.

This perspective also suggests why simply restricting the hypothesis class to a structured class is not always a satisfactory replacement. First, in the pointwise-complexity theory of Li and Xu (2026), low compression complexity depends on both the realized hypothesis and the data distribution; it is therefore a data-centric, pointwise property rather than merely membership in a fixed combinatorial subclass. Second, computationally efficient procedures may not output elements of the exact compression class. The familiar lesson from sparse regression is that tractable algorithms often work through relaxations or implicit regularization rather than by solving the exact combinatorial model-selection problem. The algorithmic lower envelope is designed for this regime: it evaluates the lower-bound obstruction through the estimator actually used, under a prior that represents the intended geometry.

Relationship to the pointwise upper theorem and existing lower bounds. The Bayesian algorithmic lower bound also clarifies the relation to the pointwise upper theorem. The upper envelope is frequentist, simultaneous, multiscale, and high-probability: it is pathwise in the Gaussian field and integrates local prior masses across scales. By contrast, the lower bound is Bayesian, one-scale, and estimator-dependent: it averages over $\Theta \sim \pi$ and $Y \sim P_\Theta$ for a fixed estimator, while the additional ghost/reference law $Y \sim Q$ enters through the quantile term $\rho_{\Delta,Q}(\widehat{t})$. This parallels the information-theoretic upper–lower duality emphasized by Zhang (2006), but with the additional point that the lower bound is a data-processing statement valid for every estimator.

DEC-type lower bounds (Foster et al., 2021, 2023) provide an independent motivation for incorporating algorithmic dependence into the fundamental limits of online regret, exploration, and interactive decision making. This perspective further motivates the Bayesian algorithmic lower bound of Chen et al. (2024) and related variants (Xu and Zeevi, 2025b). The present algorithmic lower-bound perspective isolates a distinct but complementary principle: a prior-mass lower-bound analogue of pointwise complexity, in the spirit of the pointwise generalization framework of Li and Xu (2026) for deep representation learning.

1.3 Main finite-scale renormalization theorem

We next state the finite-dimensional renormalization theorem in the form used later. Its main role is to provide an analogue of the pointwise complexity theory for deep neural networks developed in Li and Xu (2026). Let a depth-J RG trajectory be parametrized by scale variables

$$g = (g_0, \dots, g_{J-1}), \qquad g_j \in B_2(R_j) \subset \mathbb{R}^{p_j},$$

and let $S_J(g)$ denote the final effective action, or any finite-dimensional observable extracted from the final effective action, equipped with a semimetric ρ. The theorem is conditional on a finite-scale domination statement of the form

$$\rho(S_J(g'), S_J(g))^2 \le \sum_{j=0}^{J-1} (g'_j - g_j)^\top G_j(g,\varepsilon)(g'_j - g_j) \tag{1.23}$$

for all g' in a finite ρ-neighborhood of g. This defines the pointwise ellipsoidal RG metric

$$\rho_{G,g,\varepsilon}(g,g')^2 := \sum_{j=0}^{J-1} (g'_j - g_j)^\top G_j(g,\varepsilon)(g'_j - g_j),$$

which is the RG analogue of the pointwise ellipsoidal metric induced by learned feature Gram matrices in the DNN theory. The matrices $G_j(g,\varepsilon)$ are not infinitesimal Hessians; they are finite-scale secant response Grams produced by replacing one RG scale at a time. Section 3 proves (1.23) from an exact telescoping identity whenever the one-step RG replacements have secant covariance bounds and the later RG flow is locally stable.

For a positive semidefinite matrix G, radius R, and resolution $u > 0$, define

$$r_{\mathrm{eff}}(G,R,u) := \max\{k : \lambda_k(G)R^2 \ge u^2\}, \qquad d_{\mathrm{ell}}(G,R,u) := \frac{1}{2}\sum_{k=1}^{r_{\mathrm{eff}}} \log\left(\frac{CR^2\lambda_k(G)}{u^2}\right), \tag{1.24}$$

with eigenvalues in nonincreasing order. Let $V_j(g,\varepsilon)$ be the active eigenspace of $G_j(g,\varepsilon)$, and let $\mathfrak{A}_j(g,\varepsilon,u_j)$ be the atlas cost of choosing the phase label, effective rank, and a reference subspace close to $V_j(g,\varepsilon)$. A concrete Grassmannian expression is given in Section 3.

Theorem 1.8 (Finite-scale RG pointwise dimension, main form)**.** *Assume the finite-scale domination (1.23). Choose numbers $u_j > 0$ such that $\sum_j u_j^2 \le c\varepsilon^2$. Then there exists a data-independent hierarchical prior Π over phase labels, active ranks, reference subspaces, and local scale couplings such that*

$$\log \frac{1}{\Pi(B_\rho(g,\varepsilon))} \le C \sum_{j=0}^{J-1} \left[d_{\mathrm{ell}}(G_j(g,\varepsilon), R_j, u_j) + \mathfrak{A}_j(g,\varepsilon,u_j) \right]. \tag{1.25}$$

Thus finite-scale renormalization has the same complexity structure as nonlinear feature-learning DNN: a local ellipsoidal effective dimension plus a global atlas cost at each scale.

The theorem should be read as an abstract finite-cutoff theorem, not as a shortcut to continuum constructive QFT. To prove a scaling-limit theorem such as four-dimensional marginal triviality, one also needs a model-specific contraction or Gaussianization estimate for non-Gaussian connected correlations. The pointwise-dimension theorem supplies the local finite-scale complexity layer once such RG or correlation estimates are available.

1.4 Main local-time application: target fields versus cover time

The graph application distinguishes two ways of using Gaussian information. The field-level route fixes a target set, controls the pinned GFF on that target by a pointwise envelope, and then transfers the bound through Ray–Knight to a local-time statement. The global route first proves that the entire graph has been covered and only afterwards infers that the target has been visited. These two stopping-time arguments can have different scales.

Section 4.2 proves this separation on phase-star hierarchies. A target subatlas I is encoded by an atlas prior q, and the same ambient pointwise envelope can be evaluated on the selected target. The exact law

$$\mathbb{P}\{\tau_I \le \tau(t)\} = (1-e^{-t})^{|I|}, \qquad \mathbb{P}\{\tau_{\mathrm{cov}} \le \tau(t)\} = (1-e^{-t})^{N}$$

shows that target cover has intrinsic inverse-root-local-time scale $\log |I|$, whereas any route through full cover pays $\log N$. The pointwise envelope pays the target code length $\log(|I|/q(I))$, and the chart-decorated phase-star adds a compressed within-chart term. This gives a concrete finite-graph analogue of the phase/subspace atlas cost in Section 3.

1.5 Organization

Section 2 proves the pointwise Gaussian theorem and the Bayesian algorithmic lower envelopes. Section 3 develops the finite-scale renormalization pointwise-complexity theory, including the DNN–RG dictionary, secant-Gram metric domination, phase/subspace atlas covering, and the relation to marginal triviality, universal approximation, and Yang–Mills. Section 4 gives the graph and local-time applications, including target-set Ray–Knight corollaries and the phase-star separation between local-time and global cover-time scales. Section 5 summarizes the finite-cutoff scope and the main lessons.

2 Proofs of the Gaussian and Algorithmic Lower Results

2.1 Proof of the Pointwise Gaussian Theorem

The proof follows a subset-homogeneity and peeling strategy. First, for each fixed subset $H \subseteq T$, we combine an anchored form of Fernique's majorizing-measure upper bound (Fernique, 1975; Talagrand, 1987), the ambient equivalence principle of pointwise dimension, and Borell–Tsirelson concentration. Second, we apply the one-parameter uniform pointwise peeling lemma twice: first to localize the Fernique–Talagrand integral $\Phi_\mu(x)$, and then to localize the variance scale from R_* to $\sigma(x)$. This avoids introducing a separate two-dimensional peeling corollary.

The following uniform pointwise peeling lemma was originally developed for empirical processes in Xu and Zeevi (2025a). It identifies minimal conditions for obtaining pointwise envelopes, up to negligible double-logarithmic factors, without imposing unrealistic Bernstein-type or rigid sub-root assumptions. Necessary and sufficient justifications and discussions of data dependence are provided in Li and Xu (2026). Here we extend the lemma to general indexed processes, including Gaussian processes.

Lemma 2.1 (Uniform pointwise peeling for indexed processes)**.** *Let $\{Z_x\}_{x\in T}$ be random variables, let $d_0 : T \to [0, R]$, and let $\psi(r;\delta)$ be nondecreasing in r and nonincreasing in δ. Assume that for every fixed subset $H \subseteq T$ and every $\delta \in (0, 1)$,*

$$\mathbb{P}\left(\sup_{x\in H} Z_x \le \sup_{x\in H} \psi(d_0(x);\delta)\right) \ge 1-\delta. \tag{2.1}$$

Then for every $r_0 > 0$, with probability at least $1-\delta$, uniformly over all $x \in T$,

$$Z_x \le \psi\left(\max\{2d_0(x), r_0\}; \frac{\delta}{1+\lceil \log_2^+(R/r_0)\rceil}\right), \tag{2.2}$$

where $\log_2^+(u) = \max\{\log_2 u, 0\}$.

Proof. Let $m := \lceil \log_2^+(R/r_0)\rceil$. The sets

$$H_0 := \{x : d_0(x) \le r_0\}, \qquad H_j := \{x : 2^{j-1}r_0 < d_0(x) \le 2^j r_0\} \quad (1 \le j \le m)$$

cover T. Apply (2.1) to each H_j with confidence level $\delta/(m+1)$, and take a union bound. On the resulting event, if $x \in H_0$, then

$$Z_x \le \psi\left(r_0; \frac{\delta}{m+1}\right) \le \psi\left(\max\{2d_0(x), r_0\}; \frac{\delta}{m+1}\right),$$

and if $x \in H_j$ for $j \ge 1$, then $2^j r_0 \le 2d_0(x)$, so

$$Z_x \le \psi\left(2^j r_0; \frac{\delta}{m+1}\right) \le \psi\left(\max\{2d_0(x), r_0\}; \frac{\delta}{m+1}\right).$$

This proves the claim. □

We also use the following localized version of the ambient equivalence principle of pointwise dimension from Lemma 7 of Li and Xu (2026).

Lemma 2.2 (Ambient equivalence for the localized integral)**.** *Let (T, d) be a finite metric space and let $H \subseteq T$. Let $p : T \to H$ be a nearest-point selector, so that $d(y, p(y)) = \min_{h\in H} d(y, h)$, and let $\mu_H = p_\#\mu$ be the pushforward of an ambient prior $\mu \in \Delta(T)$. Then, for every $x \in H$ and every $\varepsilon > 0$,*

$$\mu_H(B_d(x, 2\varepsilon)) \ge \mu(B_d(x, \varepsilon)).$$

Consequently, for the localized functional Φ in (1.1),

$$\Phi_{\mu_H}(x) \le 2\Phi_\mu(x), \qquad x \in H. \tag{2.3}$$

Proof. If $y \in B_d(x, \varepsilon)$, then

$$d(p(y), x) \le d(p(y), y) + d(y, x) \le 2d(y, x) \le 2\varepsilon,$$

because $x \in H$ and $p(y)$ is a nearest point in H. Hence $p(B_d(x, \varepsilon)) \subseteq B_d(x, 2\varepsilon)$, which gives the ball-mass inequality. Therefore

$$\Phi_{\mu_H}(x) = \int_0^{4\sigma(x)} \sqrt{\log \frac{1}{\mu_H(B_d(x, u))}}\, du \le 2\int_0^{2\sigma(x)} \sqrt{\log \frac{1}{\mu(B_d(x, \varepsilon))}}\, d\varepsilon \le 2\Phi_\mu(x),$$

where we used the change of variables $u = 2\varepsilon$. □

Lemma 2.3 (Anchored subset majorizing-measure bound)**.** *For every fixed subset $H \subseteq T$, every ambient prior $\mu \in \Delta(T)$, and*

$$\sigma_H := \sup_{x \in H} \sigma(x),$$

there is an absolute constant $C > 0$ such that

$$\mathbb{E} \sup_{x \in H} X_x \le C \sup_{x \in H} \Phi_\mu(x) + C\sigma_H. \tag{2.4}$$

Proof. Let $H_0 := H \cup \{x_0\}$, and let $p : T \to H_0$ be a nearest-point selector. Define the probability measure

$$\bar{\mu}_H := \frac{1}{2}\delta_{x_0} + \frac{1}{2} p_{\#}\mu$$

on H_0. Fernique's majorizing-measure upper bound (Fernique, 1975; Talagrand, 1987), applied to the restricted process on H_0, gives

$$\mathbb{E} \sup_{x \in H} X_x \le C \sup_{x \in H_0} \int_0^{2\sigma_H} \sqrt{\log \frac{1}{\bar{\mu}_H(B_d(x,u))}}\, du,$$

since $\operatorname{diam}(H_0, d) \le 2\sigma_H$. For $x = x_0$, the integral is at most $2\sigma_H\sqrt{\log 2}$, because $\bar{\mu}_H\{x_0\} \ge 1/2$. For $x \in H$, split the integral at $4\sigma(x)$. On $[4\sigma(x), 2\sigma_H]$, the ball $B_d(x,u)$ contains x_0, hence has $\bar{\mu}_H$-mass at least $1/2$, so this part is bounded by $C\sigma_H$. On $[0, 4\sigma(x)]$, Lemma 2.2 and the extra factor $1/2$ in $\bar{\mu}_H$ give a bound of order $\Phi_\mu(x) + \sigma(x)$. Taking the supremum over $x \in H$ proves (2.4). $\square$

Lemma 2.4 (Subset-homogeneous Gaussian estimate)**.** *For every fixed subset $H \subseteq T$, every ambient prior $\mu \in \Delta(T)$, and every $\delta \in (0,1)$,*

$$\mathbb{P}\left(\sup_{x \in H} X_x \le C \sup_{x \in H} \Phi_\mu(x) + C\sigma_H \sqrt{\log(e/\delta)} \right) \ge 1 - \delta, \tag{2.5}$$

where $\sigma_H = \sup_{x \in H} \sigma(x)$.

Proof. By Lemma 2.3,

$$\mathbb{E} \sup_{x \in H} X_x \le C \sup_{x \in H} \Phi_\mu(x) + C\sigma_H.$$

Realize $X_x = \langle g, v_x \rangle$ as an isonormal process on a Hilbert space. The map $g \mapsto \sup_{x \in H}\langle g, v_x\rangle$ is σ_H-Lipschitz. Borell–Tsirelson concentration therefore gives the desired inequality, after absorbing the expectation term $C\sigma_H$ into $C\sigma_H\sqrt{\log(e/\delta)}$. $\square$

Proof of Theorem 1.1. Fix an arbitrary subset $H \subseteq T$. Applying Lemma 2.1 on the index set H, with $Z_x = X_x$, $d_0(x) = \Phi_\mu(x)$, $R = \Phi_{\mu,*}$, and using the subset estimate (2.5), yields the following: for every $r_0 > 0$ and every $\delta \in (0,1)$, with probability at least $1 - \delta$,

$$\forall x \in H: \qquad X_x \le C \max\{\Phi_\mu(x), r_0\} + C\sigma_H \sqrt{\log\left(\frac{eM_\Phi}{\delta}\right)}, \tag{2.6}$$

where

$$M_\Phi := 1 + \lceil \log_2^+(\Phi_{\mu,*}/r_0) \rceil.$$

Since H was arbitrary, (2.6) is now a subset-homogeneous estimate for the centered residual

$$Y_x := X_x - C\max\{\Phi_\mu(x), r_0\}$$

with complexity $\sigma(x)$. Applying Lemma 2.1 a second time, now with $Z_x = Y_x$, $d_0(x) = \sigma(x)$, $R = R_*$, and

$$\psi(s;\delta) := Cs\sqrt{\log\left(\frac{eM_\Phi}{\delta}\right)},$$

gives, with probability at least $1-\delta$,

$$\forall x \in T: \qquad Y_x \le C\max\{\sigma(x), s_0\}\sqrt{\log\left(\frac{eM_\Phi M_\sigma}{\delta}\right)},$$

where

$$M_\sigma := 1 + \lceil \log_2(R_*/s_0)\rceil.$$

Combining the last display with the definition of Y_x proves (1.4), since $M_{r_0,s_0} = M_\Phi M_\sigma$. The absolute-value estimate (1.5) follows by applying the same argument to $-X$ and taking a union bound. $\square$

Proof of Proposition 1.2. The upper bound follows from Lemma 2.3 with $H = T$, followed by optimization over μ. For the lower bound, choose x_* with $\sigma(x_*) = R_*$. Since $X_{x_0} = 0$,

$$\sup_{x\in T} X_x \ge (X_{x_*})_+, \qquad \mathbb{E}(X_{x_*})_+ = \frac{R_*}{\sqrt{2\pi}}.$$

It remains to lower-bound the optimized local functional. Let

$$\Phi_\mu^{\mathrm{full}}(x) := \int_0^{2R_*}\sqrt{\log\frac{1}{\mu(B_d(x,\varepsilon))}}\,d\varepsilon.$$

The classical majorizing-measure theorem gives

$$\inf_{\mu\in\Delta(T)}\sup_x \Phi_\mu^{\mathrm{full}}(x) \lesssim \mathbb{E}\sup_{x\in T} X_x.$$

Since $\Phi_\mu(x) \le \Phi_\mu^{\mathrm{full}}(x)$ for every x, we have

$$\inf_\mu \Phi_{\mu,*} \le \inf_\mu \sup_x \Phi_\mu^{\mathrm{full}}(x) \lesssim \mathbb{E}\sup_{x\in T} X_x.$$

Together with the radius lower bound this proves the left side of (1.6). The absolute-value statement follows from

$$\sup_x X_x \le \sup_x |X_x| \le \sup_x X_x + \sup_x(-X_x)$$

and symmetry. $\square$

Proof of Corollary 1.3. Let

$$Y := \sup_{x\in T}\frac{\big(|X_x| - C\max\{\Phi_\mu(x), r_0\}\big)_+}{\max\{\sigma(x), s_0\}}.$$

By the two-sided form of Theorem 1.1, after increasing C if necessary,

$$\mathbb{P}\left(Y > C\sqrt{\log\left(\frac{eM_{r_0,s_0}}{\delta}\right)}\right) \le \delta, \qquad \delta\in(0,1).$$

Integrating this tail gives

$$\mathbb{E}Y \le C\sqrt{\log(eM_{r_0,s_0})},$$

which proves (1.8). The pointwise bound (1.9) follows by dropping the supremum, and the global bound (1.10) follows by taking $r_0 \ge \Phi_{\mu,*}$, $s_0 = R_*$, and then letting $r_0 \downarrow \Phi_{\mu,*}$. Optimizing over μ and applying Proposition 1.2 proves (1.11). □

2.2 Proofs for the Bayesian algorithmic lower bounds

Proof of Proposition 1.4. The result is the Gaussian-process and KL specialization of the interactive Fano method of Chen et al. (2024, Theorem 2); we give a self-contained proof. Fix $\widehat{t}$, Q, and Δ, and let

$$E := \{(t, Y) : d(t, \widehat{t}(Y)) < \Delta\}.$$

Under the true joint law $P := \int \pi(dt)P_t(dY)$, write $p = P(E)$. Under the ghost law $P_0 := \pi \otimes Q$, write

$$q = P_0(E) = \rho_{\Delta,Q}(\widehat{t}).$$

By data processing,

$$\mathcal{I}_\pi(Q) = D_{\mathrm{KL}}(P\|P_0) \ge D_{\mathrm{KL}}(\mathrm{Bern}(p)\|\,\mathrm{Bern}(q)).$$

The elementary inequality

$$D_{\mathrm{KL}}(\mathrm{Bern}(p)\|\,\mathrm{Bern}(q)) \ge p\log\frac{1}{q} - \log 2$$

gives

$$p \le \frac{\mathcal{I}_\pi(Q) + \log 2}{\log(1/q)}.$$

Therefore

$$P\{d(t, \widehat{t}(Y)) \ge \Delta\} = 1 - p \ge 1 - \frac{\mathcal{I}_\pi(Q) + \log 2}{\log(1/\rho_{\Delta,Q}(\widehat{t}))},$$

with the positive part inserted to make the bound nonnegative. Multiplying by Δ and optimizing over Δ, Q proves (1.13). □

Proof of Theorem 1.5. For the Gaussian location experiment,

$$\mathcal{I}_\pi(Q_\tau) = \int_T D_{\mathrm{KL}}\big(N(v_t, \tau^2 I)\|N(\bar{v}_\pi, \tau^2 I)\big)\,\pi(dt) = \frac{1}{2\tau^2}\int_T \|v_t - \bar{v}_\pi\|^2\pi(dt) = \frac{\mathsf{V}_\pi}{4\tau^2}.$$

Applying Proposition 1.4 to $\widehat{\Theta}_{\mathrm{nn}}$ and Q_τ gives (1.15) under condition (1.14).

It remains to relate this Bayes location error to a Gaussian range. For every realization of (Θ, Z), the nearest-neighbor property gives

$$\|Y - v_{\widehat{\Theta}_{\mathrm{nn}}}\|^2 \le \|Y - v_\Theta\|^2.$$

Since $Y = v_\Theta + \tau Z$, expanding and cancelling $\tau^2\|Z\|^2$ yields

$$\|v_\Theta - v_{\widehat{\Theta}_{\mathrm{nn}}}\|^2 \le 2\tau\langle Z, v_{\widehat{\Theta}_{\mathrm{nn}}} - v_\Theta\rangle = 2\tau\big(X_{\widehat{\Theta}_{\mathrm{nn}}} - X_\Theta\big).$$

Taking expectations and using Jensen's inequality,

$$c^2\Delta^2 \le \Big(\mathbb{E}d(\Theta, \widehat{\Theta}_{\mathrm{nn}})\Big)^2 \le \mathbb{E}d(\Theta, \widehat{\Theta}_{\mathrm{nn}})^2 \le 2\tau\,\mathbb{E}\big[X_{\widehat{\Theta}_{\mathrm{nn}}} - X_\Theta\big].$$

This proves (1.16).

For the variance-normalized small-ball statement, use $\rho_{\Delta,Q_\tau}(\widehat{\Theta}_{\rm nn}) \le q_\pi(\Delta)$. If $L_\pi(\Delta)$ is above a universal constant, choosing $\tau^2 \asymp \mathsf{V}_\pi/L_\pi(\Delta)$ with a sufficiently large absolute constant makes (1.14) hold for some absolute $c \in (0,1)$. Since $X_{\widehat{\Theta}_{\rm nn}} - X_\Theta \le \sup_{s,u\in T}(X_s - X_u)$ pointwise in Z, (1.16) gives

$$\mathbb{E} \sup_{s,u\in T} (X_s - X_u) \gtrsim \Delta^2 \sqrt{\frac{L_\pi(\Delta)}{\mathsf{V}_\pi}}.$$

If $\mathsf{V}_\pi \le C_0\Delta^2$, this becomes $\Delta\sqrt{L_\pi(\Delta)}$, with a constant depending only on C_0. Optimizing over such localized priors gives (1.21). $\square$

Proof of Corollary 1.6. Apply Theorem 1.5 with π equal to the uniform distribution on S. Since S is 2Δ-separated, every Δ-ball contains at most one point of S, so

$$q_\pi(\Delta) = \sup_{a\in T} \pi(B_d(a,\Delta)) \le M^{-1}, \qquad L_\pi(\Delta) \ge \log M.$$

The localization assumption gives

$$\mathsf{V}_\pi = \iint d(s,t)^2\, \pi(ds)\pi(dt) \le 4C_0^2\Delta^2.$$

Equation (1.19) yields the claim. $\square$

Proof of Lemma 1.7. Let

$$r_* := \sup_{\mu\in\Delta(T)} \inf_{x\in T} \mu(B_d(x,\Delta)).$$

Then

$$N_{\rm frac}(T,d,\Delta) = \frac{1}{r_*}.$$

Since T is finite, von Neumann's minimax theorem gives

$$\begin{aligned} r_* &= \sup_{\mu\in\Delta(T)} \inf_{x\in T} \sum_{a\in T} \mu(a)\mathbf{1}\{a\in B_d(x,\Delta)\} \\ &= \inf_{\pi\in\Delta(T)} \sup_{\mu\in\Delta(T)} \sum_{x,a\in T} \pi(x)\mu(a)\mathbf{1}\{a\in B_d(x,\Delta)\} \\ &= \inf_{\pi\in\Delta(T)} \sup_{a\in T} \pi(B_d(a,\Delta)). \end{aligned}$$

Taking reciprocals yields (1.22). $\square$

3 Finite-Scale Renormalization and Pointwise Dimension

This section makes precise the analogy of finite-scale renormalization with nonlinear feature learning in Li and Xu (2026). The relevant correspondence is not the superficial statement that layers are scales. The correspondence is a proof mechanism: exact finite-scale telescoping, a learned or renormalized secant Gram matrix, and a hierarchical prior over local charts and a global atlas. This is the mechanism that makes the DNN pointwise theory computable, and it is also the mechanism that a finite-cutoff nonlinear RG analysis can use. The language is compatible with rigorous Wilsonian RG frameworks such as Brydges and Slade (2015); Bauerschmidt et al. (2019), but the theorem below is an abstract finite-dimensional statement.

Deep-network pointwise theory	Finite-scale RG analogue
Layer index ℓ	RG scale index j
Weights W_ℓ	Scale coupling vector g_j
Feature matrix $F_{\ell-1}(W, X)$	Renormalized operator family $O_j(S_j)$
Feature Gram $F_{\ell-1}F_{\ell-1}^\top$	Secant response Gram Γ_j^{sec}
Exact layer telescoping	Exact replacement over RG scales
Outer Lipschitz $M_{\ell\to L}$	Stability of later RG maps $M_{j\to J}$
Active feature eigenspace	Relevant/marginal active operator subspace
Grassmannian prior	Phase/subspace atlas prior
Feature-rank compression	Contraction or truncation of irrelevant directions
Kernel/NTK baseline	Free or massive GFF baseline

Relation to finite-range decomposition and hierarchical RG. The analogy with rigorous renormalization should be understood at the level of finite-scale architecture, not as an identity of objects. In the rigorous RG method of Bauerschmidt et al. (2019), a finite-range decomposition writes the covariance as $C = \sum_j C_j$ and turns Gaussian integration into progressive integration over fluctuation scales at each RG step. This is analogous to the scale-decomposition in each layer of a deep network. Our finite-scale telescoping plays the next, comparison-theoretic role: after the nonlinear RG maps have been defined, it compares two RG trajectories by replacing one scale at a time and thereby exposes a secant response Gram. It thus provides a new tool for analyzing the cumulative effect of multiple nonlinear composition steps.

The hierarchical Φ^4 model is likewise not literally a covering argument. Its nested blocks, local RG maps, and nonperturbative remainders provide a local-to-global organization of a field theory. Hierarchical covering in pointwise complexity serves the statistical analogue: a prior first chooses a phase, rank, or reference subspace, and then covers locally inside that chart. The common principle is that local finite-scale tractability must be made compatible with global uniform control. In rigorous RG the bookkeeping is dynamical, through the flow of couplings and remainders; in pointwise complexity it is geometric and Bayesian, through the local ellipsoidal dimension plus atlas cost.

All statements below are finite-dimensional and finite-cutoff. They are intended as a rigorous structural layer, not as a continuum construction of singular interacting fields.

3.1 Exact finite-scale telescoping for nonlinear RG maps

Let $\mathcal{S}_0, \ldots, \mathcal{S}_J$ be finite-dimensional normed spaces. For each scale j, let

$$\mathcal{R}_j^{g_j} : \mathcal{S}_j \to \mathcal{S}_{j+1}, \qquad g_j \in B_2(R_j) \subset \mathbb{R}^{p_j},$$

be a nonlinear coarse-graining map. Starting from a fixed $S_0 \in \mathcal{S}_0$, define the RG trajectory

$$S_{j+1}(g) := \mathcal{R}_j^{g_j}(S_j(g)), \qquad S_J(g) = \mathcal{R}_{J-1}^{g_{J-1}} \circ \cdots \circ \mathcal{R}_0^{g_0}(S_0).$$

This notation covers, for example, the exact Wilsonian identity

$$e^{-S_{j+1}(\Phi)} = \int e^{-S_j(\Phi+\psi_j)}\, d\gamma_j(\psi_j).$$

This is the finite-dimensional analogue of integrating out one fluctuation scale. Perturbative RG expands this map; here we keep the finite map itself.

For two coupling sequences g, g', define the hybrid trajectory

$$S_j^{(g',g;m)} := \mathcal{R}_{j-1}^{g'_{j-1}} \circ \cdots \circ \mathcal{R}_m^{g'_m} \circ \mathcal{R}_{m-1}^{g_{m-1}} \circ \cdots \circ \mathcal{R}_0^{g_0}(S_0),$$

with the evident convention at the endpoints. Then the following identity is exact.

Lemma 3.1 (Exact RG replacement telescoping)**.** *For every g, g',*

$$S_J(g') - S_J(g) = \sum_{j=0}^{J-1} \Big[\mathcal{R}_{J-1:j+1}^{g'} \big(\mathcal{R}_j^{g'_j}(S_j^{(j)}) \big) - \mathcal{R}_{J-1:j+1}^{g'} \big(\mathcal{R}_j^{g_j}(S_j^{(j)}) \big) \Big], \tag{3.1}$$

where $S_j^{(j)} := \mathcal{R}_{j-1}^{g_{j-1}} \circ \cdots \circ \mathcal{R}_0^{g_0}(S_0)$ and $\mathcal{R}_{J-1:j+1}^{g'} := \mathcal{R}_{J-1}^{g'_{J-1}} \circ \cdots \circ \mathcal{R}_{j+1}^{g'_{j+1}}$.

Proof. Insert the hybrid trajectories that use g up to scale $j-1$ and g' from scale j onward. The difference between two consecutive hybrids changes only the scale-j map. Summing these consecutive differences gives (3.1). □

Let ρ be a semimetric on $\mathcal{S}_J$. The next result is the RG analogue of the non-perturbative DNN feature expansion in Li and Xu (2026): the finite replacement identity induces an ellipsoidal metric whose matrices are finite-scale secant response Grams.

Theorem 3.2 (Secant-Gram metric domination for nonlinear RG)**.** *Fix g and a finite neighborhood $\mathcal{N}(g,\varepsilon)$. Assume that for each j:*

(i) the later RG flow is locally stable: for all states U, V arising in $\mathcal{N}(g,\varepsilon)$,

$$\rho\big(\mathcal{R}_{J-1:j+1}^{g'}(U), \mathcal{R}_{J-1:j+1}^{g'}(V)\big) \le M_{j\to J}(g,\varepsilon) \|U - V\|_{j+1};$$

(ii) the scale-j finite replacement is dominated by a positive semidefinite secant Gram $\Gamma_j(g,\varepsilon)$:

$$\big\| \mathcal{R}_j^{g'_j}(S_j^{(j)}) - \mathcal{R}_j^{g_j}(S_j^{(j)}) \big\|_{j+1}^2 \le (g'_j - g_j)^\top \Gamma_j(g,\varepsilon)(g'_j - g_j).$$

Then all $g' \in \mathcal{N}(g,\varepsilon)$ satisfy

$$\rho(S_J(g'), S_J(g))^2 \le \sum_{j=0}^{J-1} (g'_j - g_j)^\top G_j(g,\varepsilon)(g'_j - g_j), \qquad G_j := J\, M_{j\to J}(g,\varepsilon)^2 \Gamma_j(g,\varepsilon). \tag{3.2}$$

Proof. Apply Lemma 3.1, the triangle inequality for ρ, the local stability assumption, and the one-step secant-Gram bound. Cauchy–Schwarz gives a factor J when the J increments are summed; this factor is included in the definition of G_j. □

How the secant Gram is verified. Condition (ii) in Theorem 3.2 is a finite-dimensional secant calculation. Fix the state $S_j^{(j)}$ and set

$$F_j(g_j) := \mathcal{R}_j^{g_j}(S_j^{(j)}).$$

If F_j is continuously Frechet differentiable on the finite neighborhood under consideration and the next-scale norm is Hilbertian, then for $\Delta_j = g'_j - g_j$ and $g_{j,t} = g_j + t\Delta_j$,

$$F_j(g'_j) - F_j(g_j) = \int_0^1 DF_j(g_{j,t})[\Delta_j] dt,$$

and Jensen's inequality gives

$$\|F_j(g_j') - F_j(g_j)\|_{j+1}^2 \le \Delta_j^\top \left(\int_0^1 DF_j(g_{j,t})^* DF_j(g_{j,t}) dt \right) \Delta_j. \tag{3.3}$$

Thus condition (ii) holds with the pairwise secant Gram in parentheses, or with any positive-semidefinite Loewner envelope that dominates it uniformly over the finite neighborhood. For an exponential-family RG step

$$\mathcal{R}_j^g(S)(\Phi) = -\log \int \exp\Big\{-S(\Phi+\psi) - \sum_{a=1}^{p_j} g_a O_{j,a}(\Phi+\psi)\Big\} d\gamma_j(\psi),$$

one has

$$\partial_{g_a} \mathcal{R}_j^g(S)(\Phi) = \mathbb{E}_{j,\Phi,g}[O_{j,a}(\Phi+\psi)].$$

Consequently the secant Gram is the Gram matrix of the finite-scale response functions

$$\Gamma_{j,ab}^{\mathrm{sec}}(g,g') = \int_0^1 \langle m_{j,a}^t, m_{j,b}^t \rangle_{j+1} dt, \qquad m_{j,a}^t(\Phi) := \mathbb{E}_{j,\Phi,g_{j,t}}[O_{j,a}(\Phi+\psi)].$$

This is the precise RG analogue of the learned feature Gram matrix in the DNN pointwise metric.

Compressed response Grams. The scalar Lipschitz choice $\Gamma_j = L_j^2 I$ verifies condition (ii), but it gives no pointwise gain. A useful pointwise-complexity theorem requires an anisotropic response estimate. Let $K_{j+1\to J}$ be a positive-semidefinite quadratic form measuring the sensitivity of the later RG flow from scale $j+1$ to the terminal observable. The propagated response Gram is then

$$G_j^{\mathrm{RG}}(g,g') := \int_0^1 DF_j(g_{j,t})^* K_{j+1\to J} DF_j(g_{j,t}) dt. \tag{3.4}$$

Suppose that the coupling space decomposes as $\mathbb{R}^{p_j} = U_j \oplus U_j^\perp$. Let P_j be an orthonormal basis matrix for U_j. Assume that, uniformly over the finite neighborhood,

$$\|DF_j(\theta)P_j a\|_{K_{j+1\to J}}^2 \le a^\top A_j a \quad (a \in \mathbb{R}^{\dim U_j}), \qquad \|DF_j(\theta)v\|_{K_{j+1\to J}}^2 \le \tau_j \|v\|^2 \quad (v \in U_j^\perp).$$

Then, for every g' in the neighborhood,

$$G_j^{\mathrm{RG}}(g,g') \preceq 2P_j A_j P_j^\top + 2\tau_j I. \tag{3.5}$$

Indeed, write any direction $h \in \mathbb{R}^{p_j}$ as $h = P_j a + v$, with $v \in U_j^\perp$, and use $\|\xi+\zeta\|^2 \le 2\|\xi\|^2 + 2\|\zeta\|^2$. Integrating along the secant path gives (3.5) in the Loewner order. After diagonalizing A_j, the active eigenvalues give the local ellipsoidal effective dimension, while directions with $\tau_j R_j^2 \lesssim u_j^2$ are below finite resolution and do not contribute. In perturbative RG language, U_j is the finite-resolution span of relevant and marginal operator directions, while irrelevant operators contribute only the residual term. In a nonconvex model this compression is imposed chartwise, and the phase/subspace prior below pays the cost of selecting the chart and active subspace.

3.2 Hierarchical phase/subspace prior

For a positive semidefinite matrix $G \in \mathbb{R}^{p\times p}$, a Euclidean radius R, and a resolution $u > 0$, set

$$r_{\mathrm{eff}}(G,R,u) := \max\{k : \lambda_k(G)R^2 \geq u^2\}, \qquad V_{\mathrm{eff}}(G,R,u) := \mathrm{span}\{\text{top } r_{\mathrm{eff}} \text{ eigenvectors}\},$$

and

$$d_{\mathrm{ell}}(G,R,u) := \frac{1}{2}\sum_{k=1}^{r_{\mathrm{eff}}(G,R,u)} \log\left(\frac{CR^2\lambda_k(G)}{u^2}\right). \tag{3.6}$$

The following elementary estimate is the local-chart calculation used throughout. For subspaces $V, \bar{V} \subset \mathbb{R}^p$, write

$$\rho_{\mathrm{proj},G}(V,\bar{V}) := \|G^{1/2}(P_V - P_{\bar{V}})\|_{\mathrm{op}}.$$

Lemma 3.3 (Ellipsoidal local chart)**.** *Let $r = r_{\mathrm{eff}}(G,R,u)$, let $V = V_{\mathrm{eff}}(G,R,u)$, and let $\bar{V} \in \mathrm{Gr}(p,r)$ satisfy*

$$\rho_{\mathrm{proj},G}(V,\bar{V}) \leq \frac{u}{4R}.$$

Let $\pi_{\bar{V}}$ be the uniform law on $B_2(2R)\cap\bar{V}$. Then, after changing the universal numerical constant in the radius, uniformly over $g \in B_2(R)$,

$$\log\frac{1}{\pi_{\bar{V}}\{g' : (g'-g)^\top G(g'-g) \leq Cu^2\}} \leq C + d_{\mathrm{ell}}(G,R,u).$$

Equivalently, replacing u by a constant multiple gives the displayed local-chart bound used in Theorem 3.4.

Proof. The point is that the prior may be supported on the reference subspace $\bar{V}$ even though the center g need not belong to $\bar{V}$. Since $r = r_{\mathrm{eff}}(G,R,u)$, the component of any vector in the inactive subspace $V^\perp$ has G-length at most u on the Euclidean ball $B_2(R)$. The projection-distance assumption transfers the active subspace V to $\bar{V}$ with additional G-error at most $u/4$. Hence the G-ball centered at g, with radius enlarged by a universal constant, contains a translate inside $\bar{V}$ of an r-dimensional ellipsoid whose principal semi-axes are comparable to

$$\min\left\{R, \frac{u}{\sqrt{\lambda_k(G)}}\right\}, \qquad k = 1,\ldots,r.$$

Because the support of $\pi_{\bar{V}}$ is $B_2(2R)\cap\bar{V}$, this translated ellipsoid lies in the support up to another universal loss. The volume ratio between $B_2(2R)\cap\bar{V}$ and this ellipsoid is therefore bounded by

$$C\prod_{k=1}^{r}\left(\frac{CR\sqrt{\lambda_k(G)}}{u}\right),$$

with factors below one omitted precisely by the definition of r_{eff}. Taking logarithms gives $C + d_{\mathrm{ell}}(G,R,u)$. This is the same finite-dimensional volume-ratio argument used in the DNN local-chart lemma. □

Let $\mathcal{A}_j$ be a finite set of phase labels at scale j, with prior weights $q_j(a) > 0$. Conditional on a phase label a and rank r, let $\nu_{j,a,r}$ be a prior on the Grassmannian $\mathrm{Gr}(p_j, r)$. For $G_j = G_j(g,\varepsilon)$, define the atlas cost

$$\mathfrak{A}_j(g,\varepsilon,u_j) := \log\frac{1}{q_j(a_j(g))} + \log(1+p_j) + \log\frac{1}{\nu_{j,a_j,r_j}\big(B_{\rho_{\mathrm{proj},G_j}}(V_j, u_j/(4R_j))\big)}, \tag{3.7}$$

where $r_j = r_{\mathrm{eff}}(G_j, R_j, u_j)$, $V_j = V_{\mathrm{eff}}(G_j, R_j, u_j)$, and $a_j(g)$ is any chart containing the trajectory at scale j. If $\nu_{j,a,r}$ is Haar-uniform on $\mathrm{Gr}(p_j, r)$, the last term is bounded by the usual Grassmannian entropy; in an ellipsoidal projection metric one may use the anisotropic Grassmannian bound from the DNN pointwise theory (Li and Xu, 2026).

Theorem 3.4 (Finite-scale RG pointwise dimension)**.** *Assume the metric domination (3.2). Choose resolutions $u_j > 0$ such that $\sum_{j=0}^{J-1} u_j^2 \le c\varepsilon^2$. Construct a hierarchical prior Π by independently, at each scale j, sampling a phase label a, an effective rank r, a reference subspace $\bar{V} \in \mathrm{Gr}(p_j, r)$, and then sampling g_j uniformly in $B_2(2R_j) \cap \bar{V}$. Then, uniformly for every trajectory g,*

$$\log \frac{1}{\Pi(B_\rho(g,\varepsilon))} \le C \sum_{j=0}^{J-1} \left[d_{\mathrm{ell}}(G_j(g,\varepsilon), R_j, u_j) + \mathfrak{A}_j(g, \varepsilon, u_j) \right]. \tag{3.8}$$

Proof. For each scale choose the phase label $a_j(g)$, the active rank r_j, and a reference subspace $\bar{V}_j$ within projection distance $u_j/(4R_j)$ of V_j. Lemma 3.3 gives the local prior mass of the ellipsoidal u_j-ball inside this chart. Multiplying the scale-wise prior masses and including the phase, rank, and subspace probabilities gives the right-hand side of (3.8). Finally, if each scale perturbation lies in its ellipsoidal u_j-ball, then (3.2) and $\sum_j u_j^2 \le c\varepsilon^2$ imply $g' \in B_\rho(g, \varepsilon)$. Hence the product event is contained in the ρ-ball, and its prior mass lower-bounds $\Pi(B_\rho(g,\varepsilon))$. $\square$

Interpretation. The theorem is the RG counterpart of the DNN Riemannian-dimension calculation. The local term d_{ell} is the effective dimension of the finite-scale response Gram. The atlas term is the price of making the prior independent of the realized RG trajectory. Irrelevant directions do not contribute once their eigenvalues fall below the finite resolution u_j^2/R_j^2; relevant and marginal directions are precisely the active directions that remain in the sum.

3.3 Free Gaussian RG as the fixed-kernel baseline

Let $E = E_C \oplus E_F$ be finite-dimensional and let a centered Gaussian field have precision matrix

$$Q = \begin{pmatrix} Q_{CC} & Q_{CF} \\ Q_{FC} & Q_{FF} \end{pmatrix}, \qquad Q_{FF} \succ 0.$$

Define the Schur-complement precision

$$Q_C^{\mathrm{RG}} := Q_{CC} - Q_{CF} Q_{FF}^{-1} Q_{FC}.$$

Then marginalizing the fluctuation variables gives $\phi_C \sim N(0, (Q_C^{\mathrm{RG}})^{-1})$, and conditionally

$$\phi_F = -Q_{FF}^{-1} Q_{FC} \phi_C + \zeta, \qquad \zeta \sim N(0, Q_{FF}^{-1})$$

independently of ϕ_C. This is a fixed-kernel RG step: all pointwise complexity is determined by the Schur-complement covariance and the fluctuation covariance. It is therefore analogous to a kernel, GP, or NTK model rather than to nonlinear feature learning.

3.4 A finite nonconvex phase-atlas example

The previous theorem is abstract. We now give a concrete finite-volume chartwise example. The point is not to construct a full continuum double-well field, but to show how a globally non-log-concave finite measure can be controlled after it is decomposed into stable phase charts. Let

$E = E_C \oplus E_F$, let $\mathcal{A}$ be a finite set of phases, and let

$$\nu(d\phi_C, d\phi_F) = \sum_{a\in\mathcal{A}} w_a \nu_a(d\phi_C, d\phi_F), \qquad \nu_a \propto e^{-S_a(\phi_C,\phi_F)} \mathbf{1}_{\Omega_{C,a}\times\Omega_{F,a}} d\phi_C d\phi_F,$$

where the sets $\Omega_{C,a}, \Omega_{F,a}$ are closed convex sets with nonempty interior. The mixture may be globally non-log-concave and multimodal. Assume that, on each chart,

$$\nabla^2 S_a(\phi_C, \phi_F) \succeq H_a = \begin{pmatrix} A_a & B_a \\ B_a^\top & D_a \end{pmatrix}, \qquad D_a \succ 0, \qquad H_a^{\mathrm{RG}} := A_a - B_a D_a^{-1} B_a^\top \succ 0.$$

Theorem 3.5 (One-step nonconvex phase-atlas RG stability)**.** *Under the preceding assumptions, the coarse marginal is again a finite phase-atlas mixture*

$$\nu_C(d\phi_C) = \sum_{a\in\mathcal{A}} w_a^+ \nu_{C,a}(d\phi_C), \qquad \nu_{C,a} \propto e^{-S_a^{\mathrm{RG}}(u)} \mathbf{1}_{\Omega_{C,a}} du,$$

where the weights w_a^+ are the corresponding normalized marginal weights and

$$S_a^{\mathrm{RG}}(u) := -\log \int_{\Omega_{F,a}} e^{-S_a(u,v)} dv.$$

Moreover, S_a^{RG} is H_a^{RG}-strongly convex. Hence, for any finite family of coarse linear observables $Y_x(u) = \langle v_x, u\rangle$, the centered chart process $Y_x - \mathbb{E}_{\nu_{C,a}} Y_x$ is sub-Gaussian with canonical metric

$$d_a(x,y)^2 = \langle v_x - v_y, (H_a^{\mathrm{RG}})^{-1}(v_x - v_y)\rangle.$$

Consequently, the standard sub-Gaussian chaining and variance peeling arguments applied chart by chart, with the comparison metric induced by $(H_a^{\mathrm{RG}})^{-1}$, yields a simultaneous envelope for the mixture, with an additional phase-atlas cost $\log(1/\min_a w_a^+)$*, or* $\log|\mathcal{A}|$ *for equal weights.*

Proof. The marginal formula is the definition of integration over E_F. The Schur-complement lower bound follows by completing the square:

$$\begin{pmatrix} u \\ v \end{pmatrix}^\top H_a \begin{pmatrix} u \\ v \end{pmatrix} = u^\top H_a^{\mathrm{RG}} u + (v + D_a^{-1} B_a^\top u)^\top D_a (v + D_a^{-1} B_a^\top u).$$

Moreover,

$$H_a - \begin{pmatrix} H_a^{\mathrm{RG}} & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} B_a D_a^{-1} B_a^\top & B_a \\ B_a^\top & D_a \end{pmatrix} \succeq 0.$$

Hence $S_a(u,v) - \frac{1}{2} u^\top H_a^{\mathrm{RG}} u$ is jointly convex on the product chart. Including the indicators of the convex sets $\Omega_{C,a}$ and $\Omega_{F,a}$, the Prékopa–Leindler theorem implies that $S_a^{\mathrm{RG}}(u) - \frac{1}{2} u^\top H_a^{\mathrm{RG}} u$ is convex on $\Omega_{C,a}$. Thus the coarse chart is H_a^{RG}-strongly log-concave. Brascamp–Lieb and Herbst then give the stated sub-Gaussian bound for linear observables. The envelope follows from the usual generic-chaining upper bound for sub-Gaussian increments, followed by the same pointwise peeling used in the Gaussian proof and a union bound over $a \in \mathcal{A}$. □

Corollary 3.6 (Pure-phase charts for a finite-volume double-well Φ^4 model)**.** *Let $G = (V, E)$ be finite and*

$$S(\phi) = \frac{1}{2} \sum_{\{u,v\}\in E} c_{uv} (\phi_u - \phi_v)^2 + \lambda \sum_{u\in V} (\phi_u^2 - a^2)^2.$$

Fix $m > a/\sqrt{3}$ and define phase charts

$$\Omega_+ = \{\phi : \phi_u \geq m \ \forall u\}, \qquad \Omega_- = -\Omega_+.$$

The restrictions of $e^{-S(\phi)}d\phi$ to Ω_+ and Ω_- define two stable phase charts. A finite mixture of these chart measures is generally non-log-concave, but every chart has Hessian bounded below by

$$L_G + 4\lambda(3m^2 - a^2)I.$$

Therefore Theorem 3.5 applies to every finite decomposition of variables $V = C \sqcup F$, and one nonlinear coarse-graining step preserves a two-chart pointwise-complexity envelope for these pure-phase restrictions. To cover the full double-well Gibbs measure one would need additional charts, for instance droplet or interface charts, and corresponding chartwise Hessian bounds.

Proof. On either chart, $|\phi_u| \geq m$, and

$$\frac{d^2}{dt^2}\lambda(t^2 - a^2)^2 = 4\lambda(3t^2 - a^2) \geq 4\lambda(3m^2 - a^2) > 0.$$

Adding the graph Dirichlet Hessian gives the displayed lower bound. The result follows from Theorem 3.5. $\square$

Significance of the phase-atlas example. The finite nonconvex phase-atlas example should be read as a finite-cutoff bridge between convex concentration theory, high-dimensional geometry, and interacting-field intuition. Within a single stable phase chart, the measure is strongly log-concave: a uniform Hessian lower bound gives Brascamp–Lieb covariance domination and, through logarithmic-Sobolev/Herbst arguments, Gaussian-type concentration for linear and Lipschitz observables (Brascamp and Lieb, 1976; Bakry and Emery, 1985). A mixture of several such charts, however, is generally not log-concave and may be genuinely multimodal, so there need not be a single global convex potential or a single global Gaussian comparison. The phase-atlas decomposition restores tractability by separating these effects. Chartwise, the Schur-complement Hessian gives the local Gaussian comparison metric and hence the pointwise ellipsoidal complexity; globally, one pays an explicit atlas cost for selecting the relevant phase and active subspace.

This distinction is also natural from the viewpoint of convex geometry. Ordinary log-concavity gives powerful isoperimetric and tail estimates, and one-dimensional marginals of isotropic log-concave measures have universal sub-exponential tails (Borell, 1975; Lovasz and Vempala, 2007). It does not, however, generally imply dimension-free sub-Gaussian marginals in all directions. The stronger Gaussian-type comparison used here comes from chartwise strong log-concavity, or equivalently from a uniform curvature lower bound. The Kannan–Lovasz–Simonovits conjecture is relevant as a central benchmark for dimension-free isoperimetric and Poincare-type control of isotropic log-concave measures (Kannan et al., 1995). In the present finite-cutoff field-theoretic setting, the pointwise-complexity theorem uses this geometric principle in an explicit spectral form: the effective spectrum of the chartwise Hessian determines the local ellipsoidal dimension, while the phase atlas records the cost of global nonconvexity. This is the same structural role played by the local-chart/global-atlas construction in the DNN pointwise theory (Li and Xu, 2026), now expressed in the language of finite-cutoff interacting fields.

3.5 Relation to rigorous RG, ADC, universal approximation, and Yang–Mills

The finite-scale theorem above controls metric-ball complexity of an RG trajectory or of chartwise coarse observables. It does not by itself prove Wick factorization, Gaussianity of a scaling limit, or a mass gap. This distinction is essential.

Relation to rigorous finite-range RG. Finite-range decomposition and hierarchical RG provide model-specific mechanisms for constructing and controlling the scale-by-scale maps. In the notation of Bauerschmidt et al. (2019), the covariance decomposition and progressive integration create the RG trajectory; the perturbative coordinates and nonperturbative norms then control its flow. Our theorem starts after this structure has been exposed: it asks what pointwise ellipsoidal metric is induced by the finite-scale response and how the resulting active directions should be covered by a prior. Thus rigorous RG supplies the model-specific dynamical estimates, while the present theorem supplies the local-complexity bookkeeping once a response Gram is available.

Relation to Aizenman–Duminil-Copin. In the four-dimensional Ising and lattice $\lambda\Phi_4^4$ setting, Aizenman and Duminil-Copin (2021) prove marginal triviality by controlling deviations from Wick's rule through random currents and multiscale estimates. This is not primarily an amplitude-control problem: in the Ising case the microscopic spins are already uniformly bounded. The central issue is the Gaussianization of macroscopic observables, namely the decay of the non-Gaussian connected-correlation remainder. Thus Theorem 3.4 cannot reprove their theorem by itself. Its role is complementary: once a model-specific random-current or rigorous-RG argument establishes Gaussianization, the pointwise-complexity theorem supplies a local, finite-cutoff, and observable-sensitive envelope whose cost is carried only by active response-Gram directions and by the phase/subspace atlas.

Universal approximation is not a certificate. One might try to approximate the RG remainder by a neural network, using universal approximation theorems (Cybenko, 1989; Hornik et al., 1989). At a fixed finite cutoff this can approximate a continuous finite-dimensional RG map on a compact set. However, universal approximation alone gives neither a scale-uniform error bound nor preservation of locality, reflection positivity, correlation inequalities, or Wick-factorization errors. To obtain an ADC-type theorem through a DNN-parametrized RG ansatz, one would need certified estimates of the form

$$\|\mathcal{R}_j(S_j) - \widehat{\mathcal{R}}_j(S_j)\|_{\mathcal{N}_{j+1}} \le \eta_j, \qquad \sum_j \eta_j < \infty,$$

in a norm $\mathcal{N}_j$ that controls connected correlations, together with a marginal-flow estimate such as

$$\lambda_{j+1} \le \lambda_j - c\lambda_j^2 + C\eta_j$$

and contraction of irrelevant remainders. These are model-specific RG estimates, not consequences of approximation theory or covering numbers alone.

Yang–Mills. For Yang–Mills, the corresponding finite-cutoff program would have to be formulated directly for lattice gauge fields, rather than for scalar fields. The natural observables are gauge-invariant quantities such as plaquette variables, Wilson loops, and character observables. Chatterjee's probabilistic formulation makes this viewpoint explicit: finite lattice gauge theories are probability measures on edge configurations, Wilson loops are the basic gauge-invariant observables, and mass-gap questions can be phrased through exponential decay of Wilson-loop or plaquette correlations (Chatterjee, 2018). An exact scale-replacement argument should therefore expose a secant response Gram on the space of such gauge-invariant observables, while the atlas should be compatible with gauge orbits and with the active gauge-invariant operator subspaces selected by the finite-scale RG flow. A theorem of this type could organize the finite-cutoff complexity layer of a possible RG approach to Yang–Mills. It would still need to be combined with genuinely model-specific ingredients: construction of the continuum Euclidean field, Osterwalder–Schrader positivity and reconstruction, and a uniform exponential-clustering estimate for gauge-invariant correlations, which is the probabilistic manifestation of a mass gap (Jaffe and Witten, 2000; Chatterjee, 2018). Thus pointwise complexity should be viewed as a finite-cutoff organizing tool for the RG part of the problem, not as a standalone program to prove the Yang–Mills mass gap.

Summary. The common mechanism with nonlinear DNNs is exact finite-scale telescoping plus a response Gram plus a data-independent atlas. In DNNs this exposes learned feature-Gram compression. In nonlinear RG it exposes active relevant/marginal operator directions and separates them from below-resolution irrelevant directions. The theorem is therefore a rigorous finite-dimensional complexity layer; scaling-limit Gaussianity, nontrivial continuum interaction, and gauge-theoretic mass gaps require additional model-specific estimates.

4 Graph Local-Time Applications: Target Fields versus Cover Time

This section applies the pointwise Gaussian theorem to pinned Gaussian free fields and then, through Ray–Knight, to random-walk local times. The distinction from cover-time theory is clearest for target questions. A scalar cover-time theorem concerns the event that every vertex has been visited. The field-level Ray–Knight estimate can instead be evaluated on a target field before one passes to any full-cover event.

The phase-star examples below make this distinction quantitative. A target subatlas I is assigned a code length through an atlas prior q. The pointwise local-time route pays this target code length, plus any within-chart fluctuation cost; a full-cover route pays the cost of all N phases. Thus the graph examples make explicit the same local-chart/global-atlas principle that underlies the finite-scale ellipsoidal theory in Section 3.

4.1 Graph setup, pinned GFF, and Ray–Knight

Let $G = (V, E, c)$ be a finite connected weighted graph with symmetric conductances $c_{xy} = c_{yx} \geq 0$, where $c_{xy} > 0$ exactly when $\{x, y\} \in E$. Write

$$c_x := \sum_{y \in V} c_{xy}.$$

The weighted graph Laplacian acts on functions $f : V \to \mathbb{R}$ by

$$(Lf)(x) = \sum_y c_{xy}\big(f(x) - f(y)\big).$$

We consider the continuous-time random walk, started at a distinguished root v_0, that jumps from x to y at rate c_{xy}/c_x. Its normalized local time is

$$L_t(x) = \frac{1}{c_x} \int_0^t \mathbf{1}\{X_s = x\}\, ds, \qquad \tau(t) := \inf\{s \geq 0 : L_s(v_0) > t\}.$$

The pinned Gaussian free field with root v_0 is the centered Gaussian vector $\eta = \{\eta_x\}_{x \in V}$ such that $\eta_{v_0} = 0$ almost surely and whose covariance is the killed Green function

$$\mathbb{E}[\eta_x \eta_y] = \Gamma_{v_0}(x, y) := \mathbb{E}_x[L_{T_{v_0}}(y)],$$

where T_{v_0} is the hitting time of v_0. Equivalently, this covariance is the inverse of the pinned Laplacian on $V \setminus \{v_0\}$. Its canonical metric is

$$d(x,y) = \big(\mathbb{E}|\eta_x - \eta_y|^2\big)^{1/2} = \sqrt{R_{\mathrm{eff}}(x,y)},$$

where R_{eff} is the effective-resistance metric of the conductance network.

We use the following finite-graph form of the second Ray–Knight theorem (Dynkin, 1984). Let η' be an independent copy of the pinned GFF.

Theorem 4.1 (Second Ray–Knight isomorphism on a finite graph)**.** *With the above normalization, under the product law of the walk and the independent fields (η, η'),*

$$\Big(L_{\tau(t)}(x) + \tfrac{1}{2}\eta_x^2\Big)_{x\in V} \overset{d}{=} \Big(\tfrac{1}{2}(\eta'_x + \sqrt{2t})^2\Big)_{x\in V}. \tag{4.1}$$

In particular, $L_{\tau(t)}(v_0) = t$ almost surely.

4.2 Target local times versus cover-time reduction

Target local-time envelopes. Let $H \subseteq V$ be a target set. The proof below only requires a deterministic envelope for the pinned GFF on H. Such an envelope may be obtained either by applying Theorem 1.1 to the restricted field on $H \cup \{v_0\}$, or by applying it once to a larger ambient field and then evaluating the resulting bound on H. This distinction is useful in the phase-star examples, where one ambient atlas prior is fixed before the target subatlas is selected.

Corollary 4.2 (Target-set pointwise local-time envelope)**.** *Let $A_H(\cdot;\delta)$ be any deterministic function satisfying*

$$\mathbb{P}\big\{|\eta_x| \le A_H(x;\delta) \text{ for all } x \in H\big\} \ge 1-\delta$$

for the pinned GFF. Then, under the law of the walk, with probability at least $1-\delta$,

$$\forall x \in H: \qquad \big[t - \sqrt{2t}\,A_H(x;\delta/2)\big]_+ \le L_{\tau(t)}(x) \le t + \sqrt{2t}\,A_H(x;\delta/2) + \frac{1}{2}A_H(x;\delta/2)^2. \tag{4.2}$$

Let

$$\tau_H := \inf\{s \ge 0 : L_s(x) > 0 \text{ for every } x \in H\}$$

be the target-cover time of H. The preceding envelope gives the following target-cover and target-blanket criterion.

Corollary 4.3 (Target cover and target blanket criterion)**.** *Assume that for some $t > 0$, $H \subseteq V$, and $\theta \in (0, 1/\sqrt{2})$,*

$$\sup_{x\in H} A_H(x;\delta/2) \le \theta\sqrt{t}.$$

Then, under the law of the walk, with probability at least $1-\delta$,

$$(1-\sqrt{2}\,\theta)t \le L_{\tau(t)}(x) \le \Big(1 + \sqrt{2}\,\theta + \frac{1}{2}\theta^2\Big)t \qquad \forall x \in H.$$

In particular, $\tau_H \le \tau(t)$ on this event, and

$$\max_{u,v\in H} \frac{L_{\tau(t)}(u)}{L_{\tau(t)}(v)} \le \frac{1+\sqrt{2}\,\theta + \theta^2/2}{1-\sqrt{2}\,\theta}.$$

Proof of Corollary 4.2. Set

$$A_x := A_H(x; \delta/2), \qquad U_x := L_{\tau(t)}(x) + \frac{1}{2}\eta_x^2, \qquad V_x := \frac{1}{2}(\eta'_x + \sqrt{2t})^2, \qquad x \in H.$$

By Theorem 4.1, $(U_x)_{x\in H} \overset{d}{=} (V_x)_{x\in H}$. Define

$$D_x := \frac{1}{2}(\sqrt{2t} - A_x)_+^2, \qquad B_x := t + \sqrt{2t}\, A_x + \frac{1}{2}A_x^2.$$

By the envelope assumption applied to the independent copy η', the event $\{|\eta'_x| \le A_x \text{ for all } x \in H\}$ has probability at least $1-\delta/2$. On this event, $D_x \le V_x \le B_x$ for all $x \in H$. Equality in distribution therefore gives

$$\mathbb{P}\{D_x \le U_x \le B_x \text{ for all } x \in H\} \ge 1 - \delta/2.$$

Applying the same envelope to η itself and taking a union bound, with probability at least $1 - \delta$, we also have $|\eta_x| \le A_x$ for all $x \in H$. On the intersection of these events,

$$L_{\tau(t)}(x) = U_x - \frac{1}{2}\eta_x^2 \le B_x,$$

and

$$L_{\tau(t)}(x) \ge D_x - \frac{1}{2}A_x^2.$$

Using nonnegativity of local time and the identity

$$\max\left\{0, \frac{1}{2}(\sqrt{2t} - A_x)_+^2 - \frac{1}{2}A_x^2\right\} = \left[t - \sqrt{2t}\, A_x\right]_+$$

gives (4.2). The event in the conclusion depends only on the walk, so the same probability bound holds under the walk law. □

Proof of Corollary 4.3. Under the displayed small-envelope assumption, Corollary 4.2 gives, simultaneously for all $x \in H$,

$$L_{\tau(t)}(x) \ge (1 - \sqrt{2}\,\theta)t > 0$$

and

$$L_{\tau(t)}(x) \le \Big(1 + \sqrt{2}\,\theta + \frac{1}{2}\theta^2\Big)t.$$

Thus every vertex in H has been visited by time $\tau(t)$, and the displayed ratio bound follows by dividing the upper estimate by the lower estimate. □

Phase-star hierarchies. We now give a finite graph model that separates direct target-local-time control from a reduction through full cover time. If a deterministic target set is fixed in advance and one is allowed to redo the entire Gaussian analysis on that target alone, then classical generic chaining on the restricted field has the correct target scale. The point here is different. We fix one ambient atlas prior before the target is selected, and then evaluate the same pointwise envelope on any target subatlas. The target scale includes the code length of the selected subatlas under this prior. A proof that first establishes full cover loses this target dependence, because it must cover every phase.

Theorem 4.4 (Phase-star target subatlas versus full cover)**.** *Let S_N be the star graph with root v_0, leaves $\{1,\dots,N\}$, and unit conductances. Let $\mathcal{I}$ be a collection of nonempty subsets of $\{1,\dots,N\}$, and let $q \in \Delta(\mathcal{I})$. For $I \in \mathcal{I}$, set*

$$H_I := I, \qquad \tau_I := \inf\{s \geq 0 : L_s(i) > 0 \text{ for every } i \in I\}.$$

Let τ_{cov} be the cover time of the whole star. Define the ambient prior

$$\mu_q := \frac{1}{2}\delta_{v_0} + \frac{1}{4}\mathrm{Unif}\{1,\dots,N\} + \frac{1}{4}\sum_{I\in\mathcal{I}} q(I)\,\mathrm{Unif}(I). \tag{4.3}$$

Let $A_q(i;\delta)$ be the two-sided pointwise Gaussian envelope from (1.5) applied to the full pinned GFF on S_N, with prior μ_q, with $r_0 = s_0 = 1$, and with constants enlarged if necessary. Then the following hold.

(i) *The pinned GFF satisfies $\eta_1,\dots,\eta_N$ independent $N(0,1)$. Moreover, with probability at least $1-\delta$, simultaneously for all $I \in \mathcal{I}$,*

$$\sup_{i\in I} |\eta_i| \leq C\left[\sqrt{\log\frac{4|I|}{q(I)}} + \sqrt{\log\left(\frac{2+\log\log(N+3)}{\delta}\right)}\right]. \tag{4.4}$$

Consequently, the same deterministic right-hand side may be used as $\sup_{i\in I} A_q(i;\delta)$ in the Ray–Knight local-time envelope.

(ii) *For every $\theta \in (0, 1/\sqrt{2})$, there is $C_\theta < \infty$ such that, for each $I \in \mathcal{I}$,*

$$t \geq C_\theta\left[\log\frac{4|I|}{q(I)} + \log\left(\frac{2+\log\log(N+3)}{\delta}\right)\right] \quad\Longrightarrow\quad \mathbb{P}\{\tau_I \leq \tau(t)\} \geq 1-\delta. \tag{4.5}$$

The probability statement is for the selected target I. The prior μ_q and the Gaussian envelope are fixed independently of that selection.

(iii) *The exact inverse-root-local-time laws are*

$$\mathbb{P}\{\tau_I \leq \tau(t)\} = (1-e^{-t})^{|I|}, \qquad \mathbb{P}\{\tau_{\mathrm{cov}} \leq \tau(t)\} = (1-e^{-t})^{N}. \tag{4.6}$$

Thus the $(1-\delta)$-quantile for covering k prescribed leaves is

$$q_k(\delta) := -\log\left(1-(1-\delta)^{1/k}\right) = \log k + O_\delta(1). \tag{4.7}$$

In particular, $q_N(\delta) - q_{|I|}(\delta) = \log(N/|I|) + O_\delta(1)$. Hence, whenever

$$\log\frac{|I|}{q(I)} = o(\log N),$$

the direct target-local-time route has an inverse-root-local-time scale asymptotically smaller than the scale required by any argument that certifies the same target event only through full cover.

Proof. For the star graph, the pinned Laplacian on the leaves is the identity matrix, so the pinned GFF has covariance I_N. For a leaf i, $\sigma(i) = d(i, v_0) = 1$, and the star metric satisfies $d(i,j) = \sqrt{2}$ for distinct leaves. If $i \in I$, then (4.3) gives

$$\mu_q(\{i\}) \geq \frac{q(I)}{4|I|}.$$

Since the metric diameter of the star leaves is bounded by an absolute constant, the localized Fernique–Talagrand functional satisfies

$$\Phi_{\mu_q}(i) \leq C\sqrt{\log \frac{4|I|}{q(I)}} \qquad (i \in I).$$

The uniform component gives $\mu_q(\{i\}) \geq 1/(4N)$ for every leaf. Hence $\Phi_{\mu_q,*} \leq C\sqrt{\log(N+1)}$. With $r_0 = s_0 = 1$, the peeling multiplicity in Theorem 1.1 is bounded by a constant times $2+\log\log(N+3)$. The two-sided pointwise Gaussian theorem gives (4.4). Applying Corollary 4.3, with the usual confidence split, gives (4.5) after increasing C_θ.

It remains to prove (4.6). During visits to the root, the walk waits an exponential time of rate one before departing and then chooses one of the N leaves uniformly. Since $c_{v_0} = N$, accumulated root occupation time at $\tau(t)$ is Nt. Therefore, per unit normalized root local time, departures from the root form a Poisson process of rate N with independent uniform leaf marks. By Poisson thinning, for each leaf i, the number of excursions from the root to i before $\tau(t)$ is Poisson with mean t, and these counts are independent over leaves. A leaf has positive local time by $\tau(t)$ if and only if it has been chosen at least once. This gives the two identities in (4.6). Solving $(1-e^{-t})^k \geq 1-\delta$ gives (4.7), and the asymptotic formula follows from $(1-\delta)^{1/k} = 1 + k^{-1}\log(1-\delta) + o(k^{-1})$. □

The collapsed star has no within-phase geometry. The next theorem adds a local chart to each phase. It is a graph analogue of a phase/subspace atlas: the prior first pays for selecting a target subatlas, and the effective resistance inside each dense chart suppresses the within-chart fluctuation.

Theorem 4.5 (Decorated phase-star subatlas theorem)**.** *Fix $N, K \geq 1$ and $\kappa > 0$ with $\kappa(K+1) \geq 1$. Let $G_{N,K,\kappa}$ be the conductance graph with root v_0, phase centers $c_1, \dots, c_N$, and local chart vertices $z_{a,1}, \dots, z_{a,K}$ for each phase a. The root is connected to each c_a by an edge of conductance one. For each a,*

$$C_a := \{c_a, z_{a,1}, \dots, z_{a,K}\}$$

is a complete graph with edge conductance κ, and there are no other edges. Let $\mathcal{I}$ be a collection of nonempty subsets of $\{1, \dots, N\}$, and let $q \in \Delta(\mathcal{I})$. For $I \in \mathcal{I}$, define the decorated target

$$H_{I,K} := \bigcup_{a \in I} C_a.$$

Define the ambient prior

$$\mu_{q,K} := \frac{1}{2}\delta_{v_0} + \frac{1}{4}\mathrm{Unif}\left(\bigcup_{a=1}^{N} C_a\right) + \frac{1}{4}\sum_{I \in \mathcal{I}} q(I)\,\mathrm{Unif}\left(\bigcup_{a \in I} C_a\right). \tag{4.8}$$

Then the following hold.

(i) *The effective resistance within a phase is compressed:*

$$R_{\text{eff}}(x,y) \le \frac{2}{\kappa(K+1)}, \qquad x,y \in C_a, \tag{4.9}$$

while

$$R_{\text{eff}}(v_0,c_a) = 1, \qquad R_{\text{eff}}(v_0,x) \le 1 + \frac{2}{\kappa(K+1)}, \quad x \in C_a. \tag{4.10}$$

(ii) *Let $A_{q,K}(x;\delta)$ be the two-sided pointwise Gaussian envelope from (1.5) applied to the full pinned GFF on $G_{N,K,\kappa}$, with prior $\mu_{q,K}$, with $r_0 = s_0 = 1$, and with constants enlarged if necessary. Then, with probability at least $1-\delta$, simultaneously for all $I \in \mathcal{I}$,*

$$\begin{aligned}\sup_{x\in H_{I,K}} |\eta_x| \le C\Bigg[&\sqrt{\log\frac{4|I|}{q(I)}} + \sqrt{\frac{1}{\kappa(K+1)}\log\frac{4|I|(K+1)}{q(I)}} \\ &+ \sqrt{\log\left(\frac{2+\log\log((N+3)(K+3))}{\delta}\right)}\Bigg].\end{aligned} \tag{4.11}$$

Consequently, for every $\theta \in (0,1/\sqrt{2})$, there is $C_\theta < \infty$ such that, for the selected I,

$$\begin{aligned}t \ge C_\theta\Bigg[&\log\frac{4|I|}{q(I)} + \frac{1}{\kappa(K+1)}\log\frac{4|I|(K+1)}{q(I)} \\ &+ \log\left(\frac{2+\log\log((N+3)(K+3))}{\delta}\right)\Bigg].\end{aligned} \tag{4.12}$$

implies

$$\mathbb{P}\{\tau_{H_{I,K}} \le \tau(t)\} \ge 1-\delta.$$

(iii) *Let τ_{cov} be the cover time of the full decorated graph, and set $C_N^{\text{cen}} := \{c_1,\ldots,c_N\}$.*

$$\mathbb{P}\{C_N^{\text{cen}} \subseteq X[0,\tau(t)]\} = (1-e^{-t})^N. \tag{4.13}$$

Since full cover implies that all phase centers have been visited,

$$\mathbb{P}\{\tau_{\text{cov}} \le \tau(t)\} \le (1-e^{-t})^N. \tag{4.14}$$

Thus any argument that proves target cover only by first proving full cover at confidence $1-\delta$ must use inverse-root-local-time at least $q_N(\delta)$ from (4.7). If, for fixed $\delta \in (0,1)$,

$$\log\frac{|I|}{q(I)} + \frac{1}{\kappa(K+1)}\log\frac{|I|(K+1)}{q(I)} + \log\log((N+3)(K+3)) = o(\log N),$$

then the sufficient scale in (4.12) is $o(q_N(\delta))$, while $\mathbb{P}\{\tau_{\text{cov}} \le \tau(t)\} \to 0$ along that scale.

Proof. The resistance estimate inside C_a is the standard effective resistance of a complete graph with $K+1$ vertices and edge conductance κ, namely $2/[\kappa(K+1)]$. Since c_a is the only vertex of C_a connected to the root, attaching the remaining clique to c_a does not change the effective resistance between v_0 and c_a; hence $R_{\text{eff}}(v_0,c_a) = 1$. The triangle inequality for effective resistance gives (4.10).

Put

$$\alpha_{K,\kappa} := \sqrt{\frac{2}{\kappa(K+1)}}.$$

For $x \in C_a$ and $a \in I$, the ball $B_d(x, \alpha_{K,\kappa})$ contains C_a by (4.9). Under $\mu_{q,K}$, this ball has mass at least $q(I)/(4|I|)$. For smaller radii, the singleton mass is at least $q(I)/[4|I|(K+1)]$. Also $\sigma(x) \le 2$ by (4.10). Therefore, for $x \in C_a$ with $a \in I$,

$$\Phi_{\mu_{q,K}}(x) \le C\sqrt{\log \frac{4|I|}{q(I)}} + C\alpha_{K,\kappa}\sqrt{\log \frac{4|I|(K+1)}{q(I)}}.$$

The uniform component in $\mu_{q,K}$ gives singleton mass at least $1/[4N(K+1)]$, and hence $\Phi_{\mu_{q,K},*} \le C\sqrt{\log((N+1)(K+1))}$. With $r_0 = s_0 = 1$, the peeling multiplicity contributes only the displayed $\log\log((N+3)(K+3))$ term. This proves (4.11). The target-cover implication follows from Corollary 4.3 as in the proof of Theorem 4.4.

Finally, each departure from the root chooses one of the phase centers uniformly, and the same Poisson-thinning argument used in Theorem 4.4 gives (4.13). Since full cover requires every phase center to have been visited, (4.14) follows. The final asymptotic statement follows from $q_N(\delta) = \log N + O_\delta(1)$. □

Relation to classical cover-time theory. For a finite connected conductance graph $G = (V, E, c)$, let τ_{cov} be the first time every vertex has been visited, and write

$$t_{\mathrm{cov}}(G) := \max_{v\in V} \mathbb{E}_v[\tau_{\mathrm{cov}}]. \tag{4.15}$$

A blanket time strengthens cover time by requiring that the walk has accumulated roughly stationary proportions of visits at all vertices. One convenient strong version leads to

$$t_{\mathrm{bl}}(G,\delta) := \max_{v\in V} \mathbb{E}_v[\tau_{\mathrm{bl}}(\delta)]. \tag{4.16}$$

Let $H(u,v)$ be the expected hitting time from u to v, and define the commute-time metric

$$\kappa(u,v) := H(u,v) + H(v,u).$$

Let $\mathcal{C} := \sum_{x\in V} c_x$ be the total conductance. The commute-time identity gives

$$\kappa(u,v) = \mathcal{C}\, R_{\mathrm{eff}}(u,v).$$

For an unweighted graph, $\mathcal{C} = 2|E|$. If $\eta = \{\eta_v\}_{v\in V}$ denotes the pinned GFF, then $\mathbb{E}(\eta_u - \eta_v)^2 = R_{\mathrm{eff}}(u,v)$.

Theorem 4.6 (Ding–Lee–Peres (Ding et al., 2012))**.** *For any connected conductance graph $G = (V, E, c)$ and any fixed $0 < \delta < 1$,*

$$t_{\mathrm{cov}}(G) \;\asymp\; \gamma_2\big(V, \sqrt{\kappa}\big)^2 \;=\; \mathcal{C}\cdot \gamma_2\big(V, \sqrt{R_{\mathrm{eff}}}\big)^2 \;\asymp_\delta\; t_{\mathrm{bl}}(G,\delta). \tag{4.17}$$

Equivalently,

$$t_{\mathrm{cov}}(G) \;\asymp\; \mathcal{C}\cdot\Big(\mathbb{E}\max_{v\in V}\eta_v\Big)^2. \tag{4.18}$$

Theorem 4.6 is a global theorem: it identifies the scale of the time needed to cover every vertex. The target local-time estimates above answer a different question. They keep a single ambient field-level envelope and evaluate it on a selected subatlas before any full-cover event is invoked. Theorems 4.4 and 4.5 show that this distinction can be strict. On the star, a selected target I has inverse-root-local-time quantile $\log|I| + O_\delta(1)$, while full cover has quantile $\log N + O_\delta(1)$. The pointwise target envelope pays the atlas code length $\log(|I|/q(I))$; in the decorated star it also pays the compressed within-chart term $(\kappa(K+1))^{-1}\log(|I|(K+1)/q(I))$. Any argument that first passes through the full-cover event necessarily pays the coupon-collector cost associated with all N phase centers. If a target set is fixed in advance, and the Gaussian analysis is allowed to be restricted to that target, then classical generic chaining applied to the restricted field can recover the corresponding target scale. The purpose of Theorems 4.4 and 4.5, however, is different: they construct a simultaneous envelope valid uniformly over all target subsets I. The pointwise formulation is what makes this possible. It retains the dependence on the selected subatlas under a single ambient prior, rather than collapsing the analysis at the outset to the full-cover event.

5 Scope and Outlook

The paper has four linked finite-scale messages. First, for Gaussian processes, the natural refinement of generic chaining is a simultaneous pointwise envelope for the field, not only a scalar estimate on $\mathbb{E}\sup_x X_x$. This envelope is sharp in expectation after optimizing over the prior and recovers the anchored Fernique–Talagrand scale. It should be useful whenever one wants to retain local field information before taking a final supremum.

Second, the Bayesian algorithmic lower envelope gives a one-scale lower-bound counterpart to pointwise complexity. It is not itself a lower bound on a Gaussian supremum. It is an information-theoretic Bayes-risk bound for every estimator $\widehat{t}(Y)$, expressed through the exact ghost small-ball mass $\mathbb{E}_Q\pi(B_d(\widehat{t}(Y),\Delta))$. The Gaussian nearest-neighbor comparison turns this Bayes distance obstruction into a lower bound on a Gaussian range, and the localized hard-prior spreading relaxation recovers a Sudakov-type form. The unrestricted classical Sudakov minoration requires an additional standard Sudakov comparison or multiscale localization step. Keeping the algorithmic mass is the pointwise refinement: the cost is attached to the estimator's realized image rather than to a class-wide worst case.

Third, finite-cutoff renormalization illustrates the same structural mechanism in a nonlinear setting. Exact telescoping exposes a response Gram, and a hierarchical atlas converts trajectory-dependent active directions into a data-independent prior. The RG theorem is finite-dimensional and conditional on model-specific response estimates; continuum interacting QFT, four-dimensional marginal triviality, and Yang–Mills require substantial contraction, Gaussianization, or gauge-invariant correlation estimates.

Fourth, the graph local-time application shows how the Gaussian field envelope is used before reducing to cover time. Ray–Knight transfers pointwise GFF control to target-set local-time control, and the phase-star examples demonstrate that a selected subatlas can have a smaller intrinsic scale than any route through full cover. Across these settings, the contribution is the same finite-scale complexity layer: expose the correct ellipsoidal metric, keep the relevant local prior mass, and pay the atlas cost only when global uniformity requires it.

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